

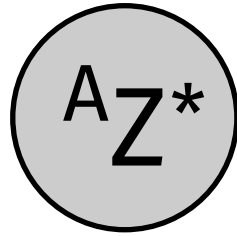
Detailed Background Characterization for Multi-Particle Correlation Spectra

Bryan M. Harvey

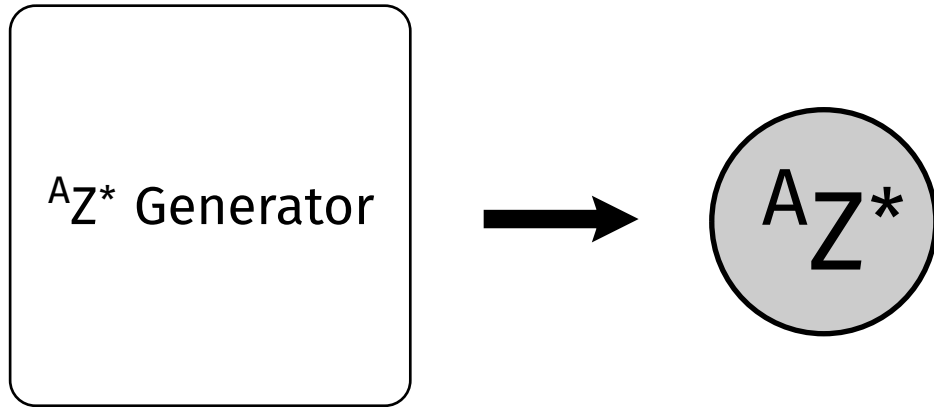
ARUNA Seminar Series

25 Sep 2026

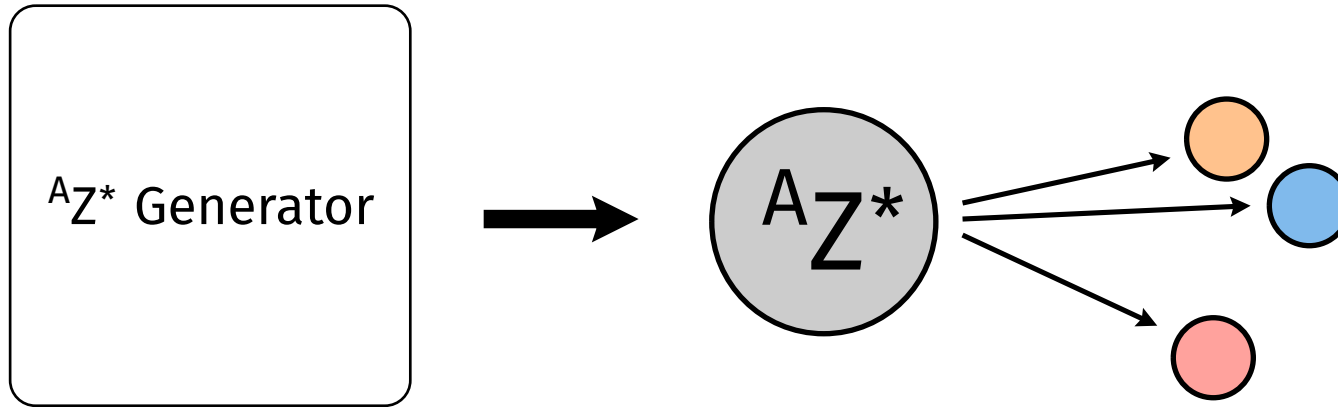
Multi-particle Correlations as a Tool



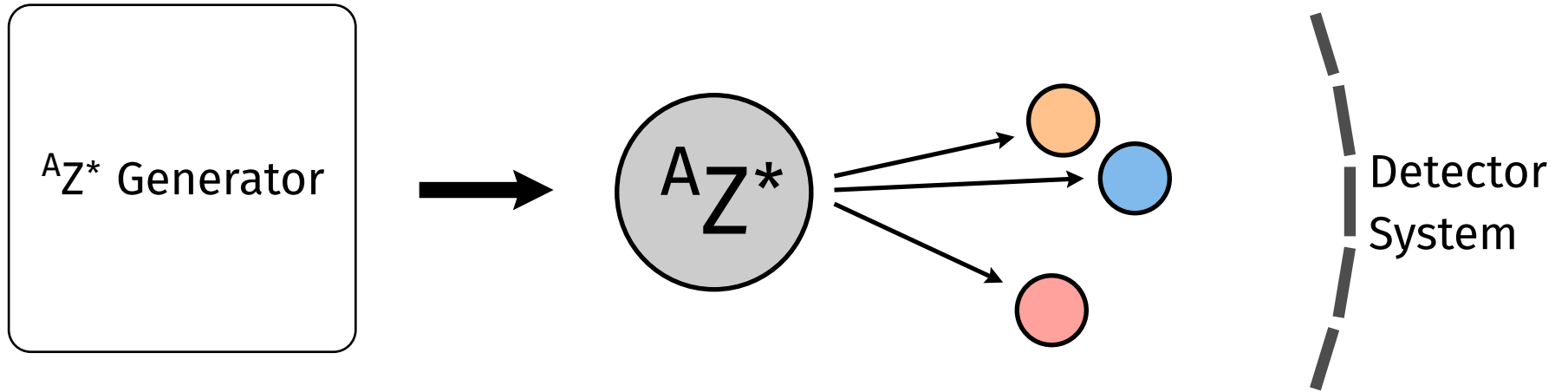
Multi-particle Correlations as a Tool



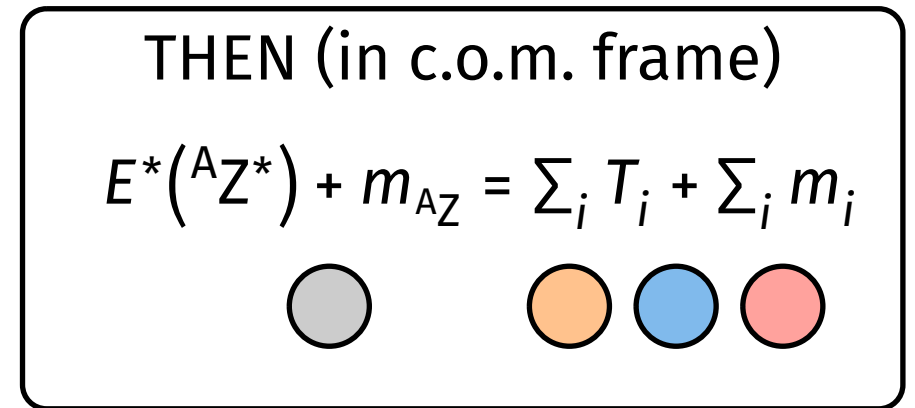
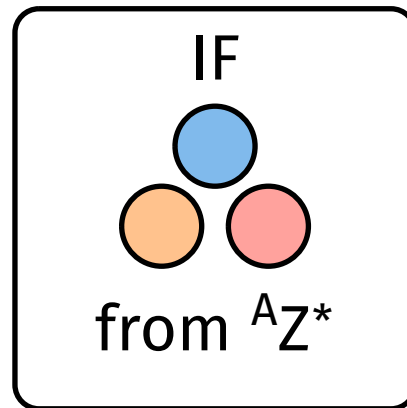
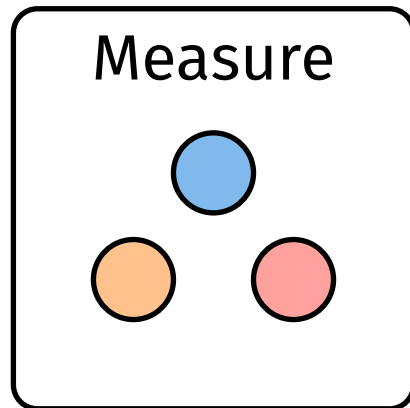
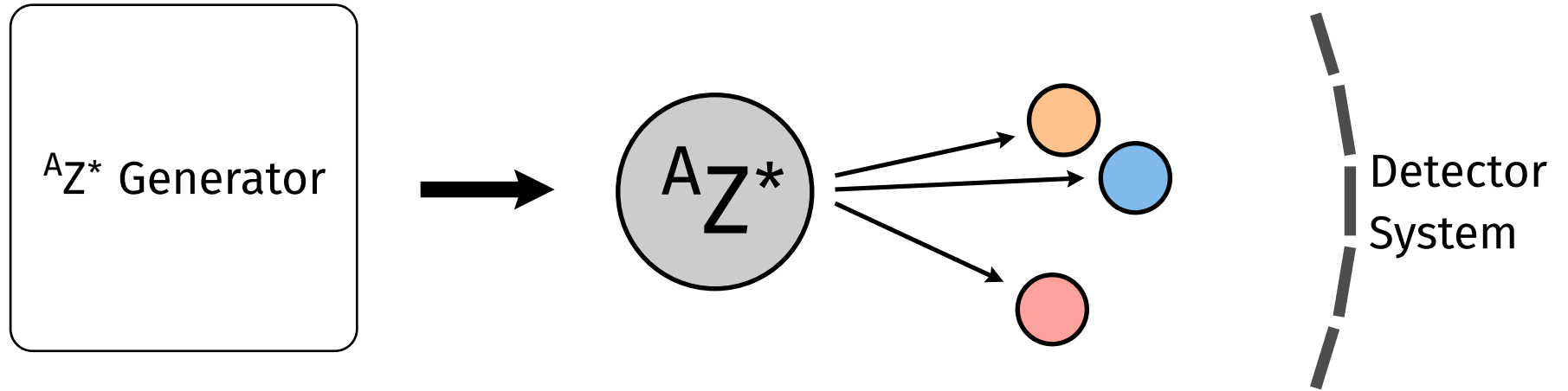
Multi-particle Correlations as a Tool



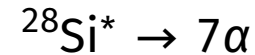
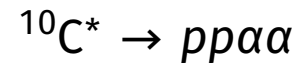
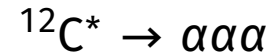
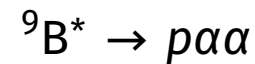
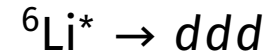
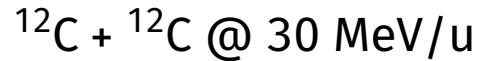
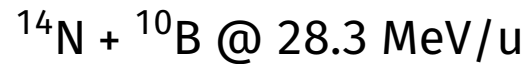
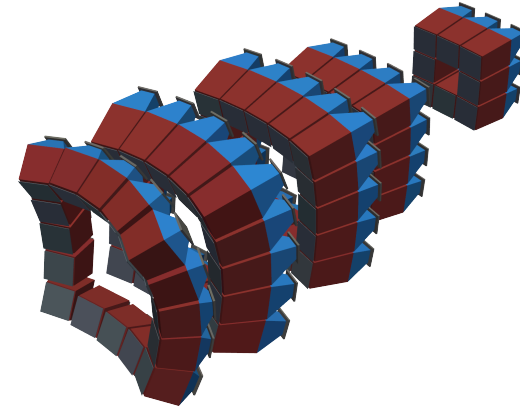
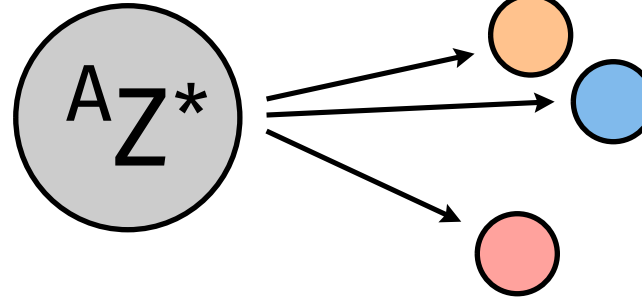
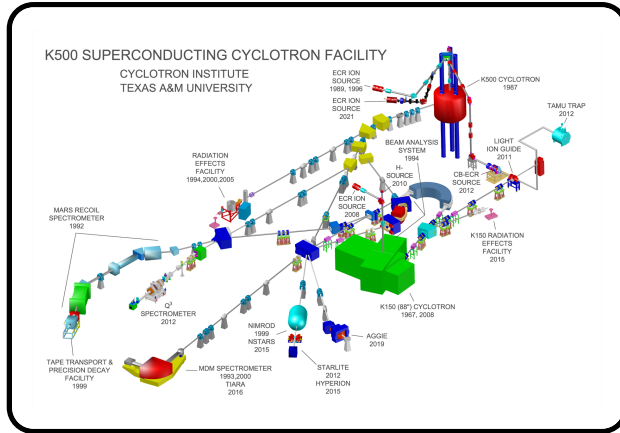
Multi-particle Correlations as a Tool



Multi-particle Correlations as a Tool

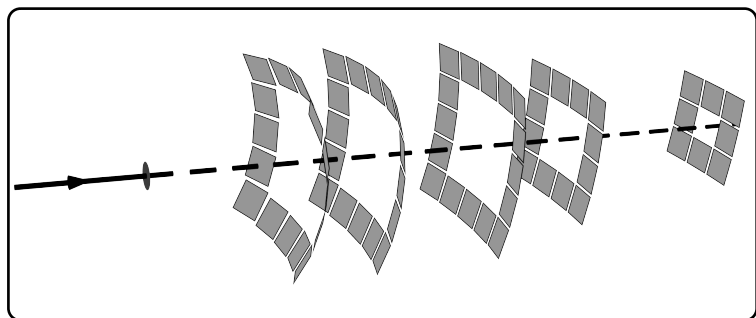
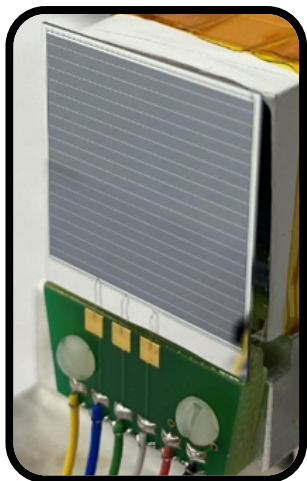
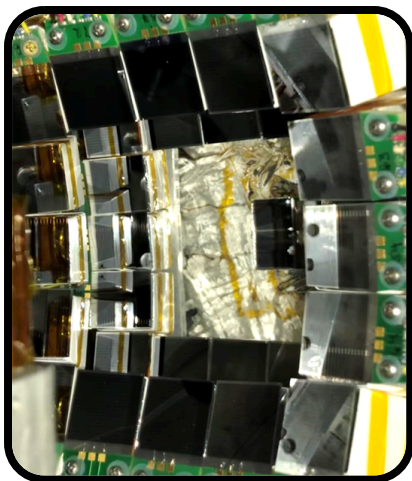


Multi-particle Correlations as a Tool



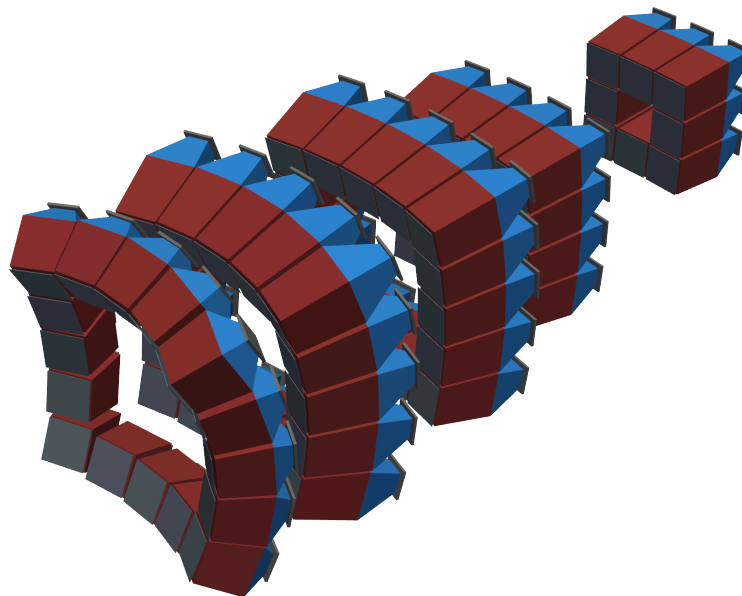
**Forward
Array
Using
Silicon
Technology**

Forward Array Using Silicon Technology (FAUST)



Forward Array Using Silicon Technology (FAUST)

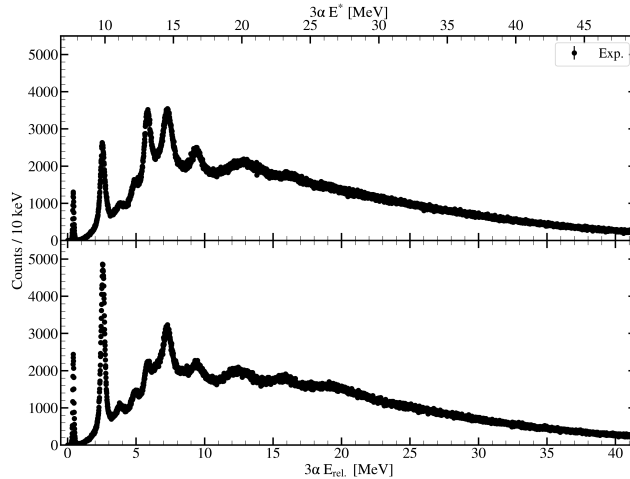
- 68 Si-CsI(Tl) telescopes
 - Excellent forward coverage ($1.7^\circ - \sim 40^\circ$)
 - Lab energy (Thick CsI stops particles)
 - Isotopic particle identification ($E-\Delta E$ technique)
 - Lab angle (Dual-Axis Duo-Lateral detectors)



Two Active Studies

Some states of $^{12}\text{C}^*$ are observed to decay into 3α .

How does the population of these states change for two similar reactions?

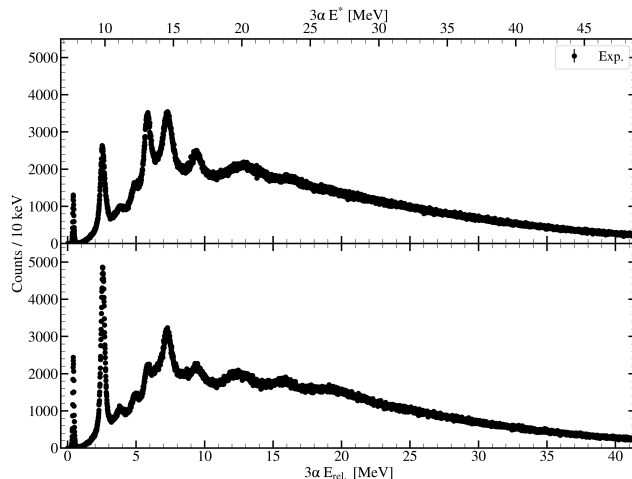


Both studies (and many more) need a detailed characterization of the nonresonant background.

Two Active Studies

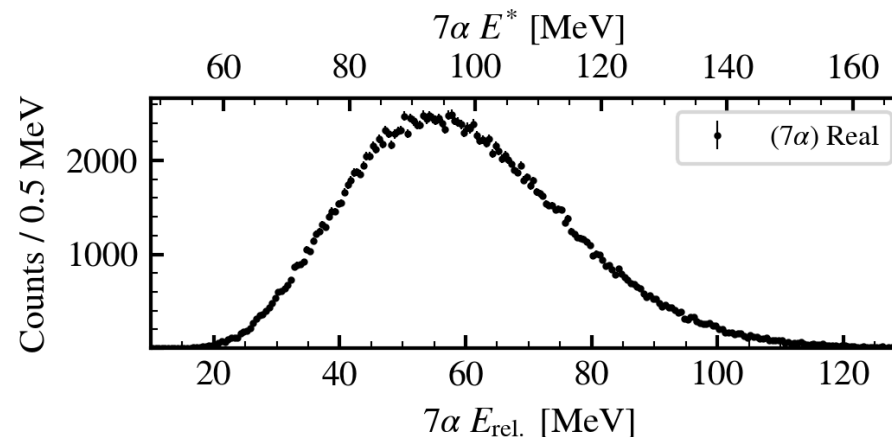
Some states of $^{12}\text{C}^*$ are observed to decay into 3α .

How does the population of these states change for two similar reactions?



Predictions for exotic α -clustered structure in $^{28}\text{Si}^*$ have been predicted.

Is there any evidence of $^{28}\text{Si}^* \rightarrow 7\alpha$ resonant decays?



Both studies (and many more) need a detailed characterization of the nonresonant background.

Characterizing the Nonresonant Background

A Case Study on *d-d-d*

- Any signals should be **rare to nonexistent**
- $Q(^6\text{Li} \rightarrow 3d) = -25.3 \text{ MeV}$
- $Q(\alpha \rightarrow 2d) = -23.8 \text{ MeV}$
- $Q(d \rightarrow pn) = -2.2 \text{ MeV}$

$$^6\text{Li}(E^* > 25.32 \text{ MeV})$$

$$\rightarrow \alpha^*(E^* > 23.8 \text{ MeV})$$

$$+ d(E^* < 2.2 \text{ MeV})$$

$$\rightarrow 3d(E^* < 2.2 \text{ MeV})$$

$$\text{Total} = \text{Background} + \text{Signal}$$

A Case Study on d - d - d

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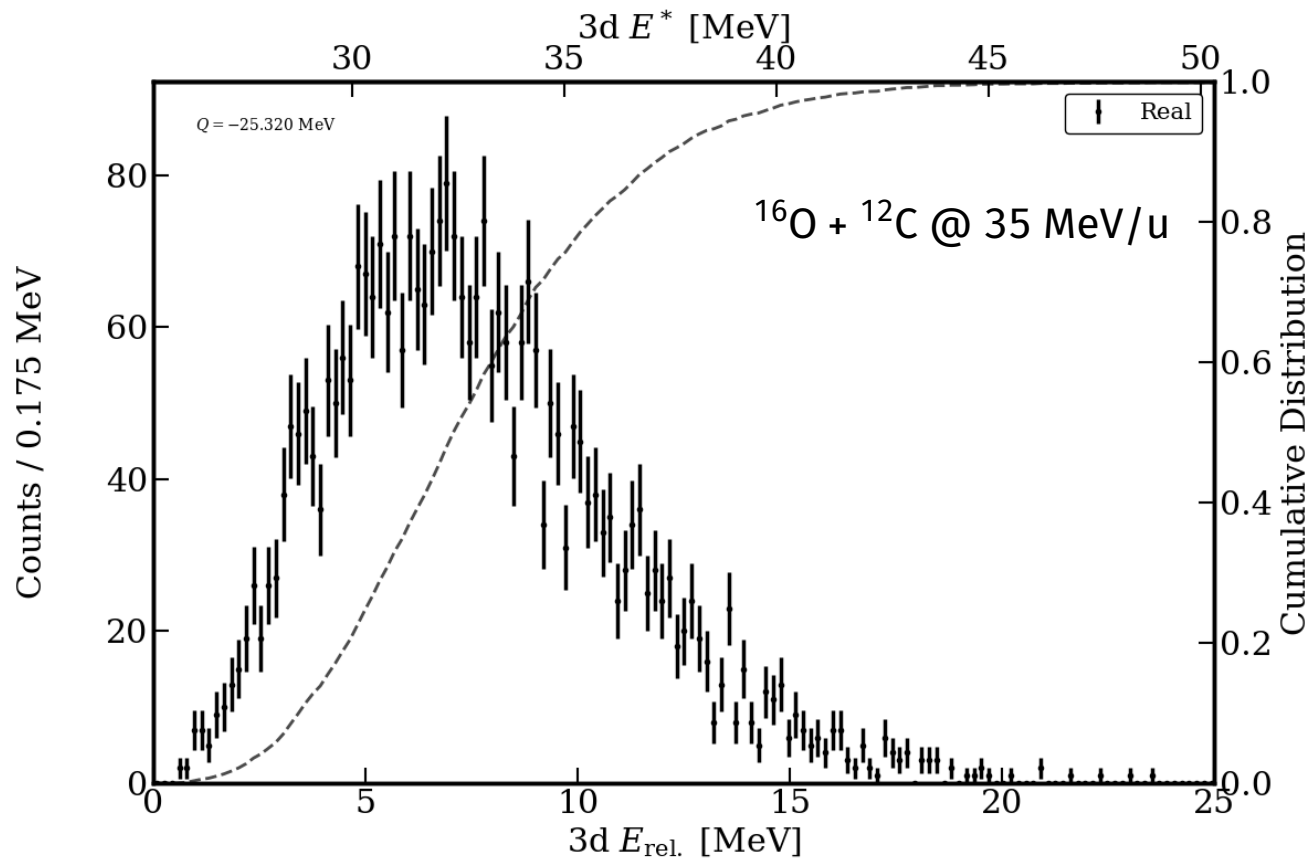
$^6\text{Li}(E^* > 25.32 \text{ MeV})$

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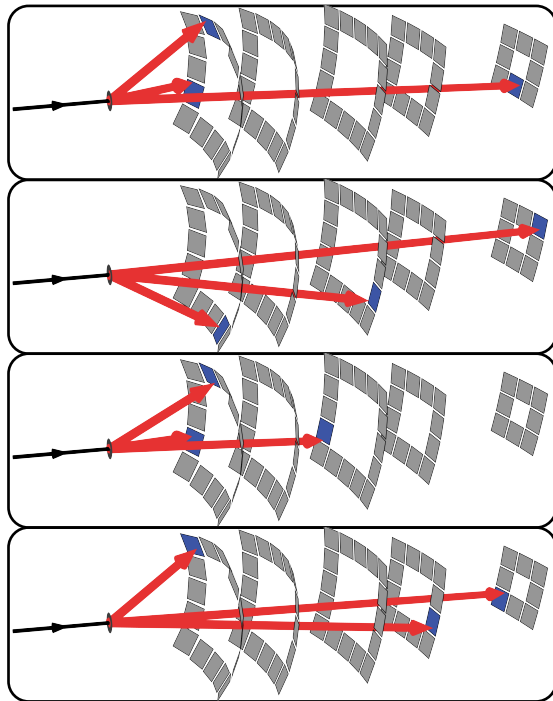
$\rightarrow 3d(E^* < 2.2 \text{ MeV})$

Total = Background + Signal



Creating a Mixed Event

Real Events

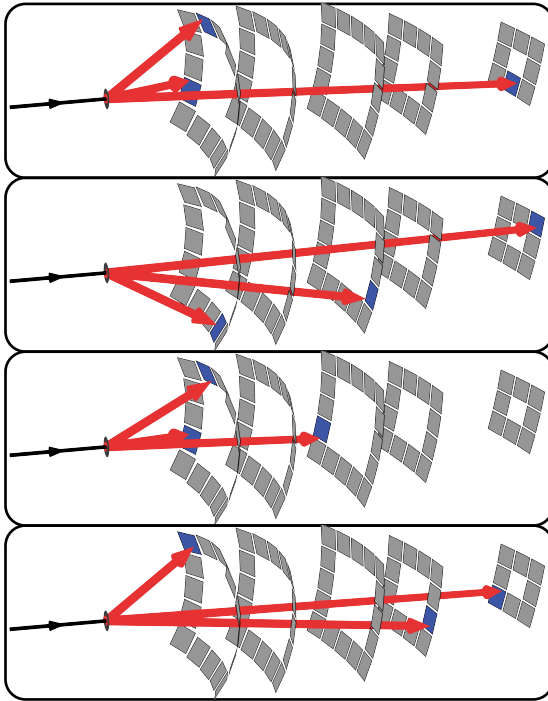


⋮

- Start w/ events containing all particles

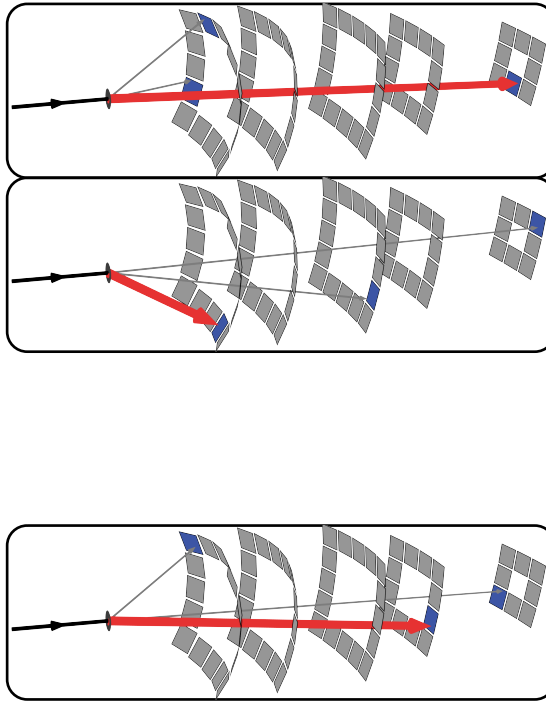
Creating a Mixed Event

Real Events



⋮

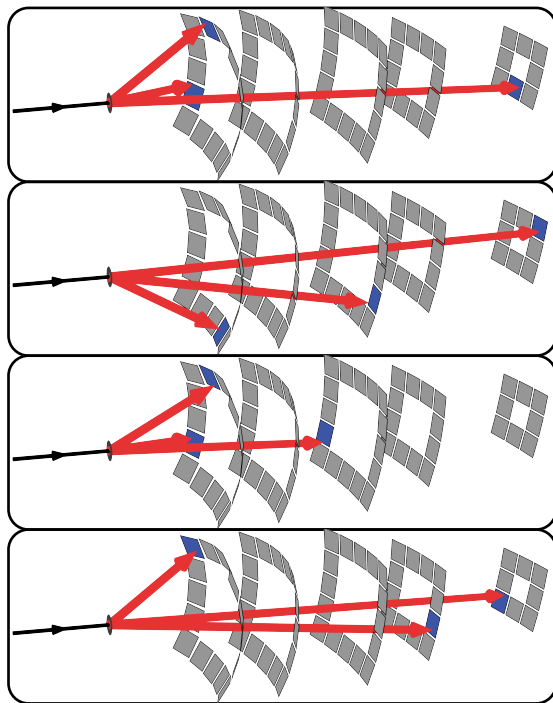
Selected Events



- Start w/ events containing all particles
- Select particles from separate events

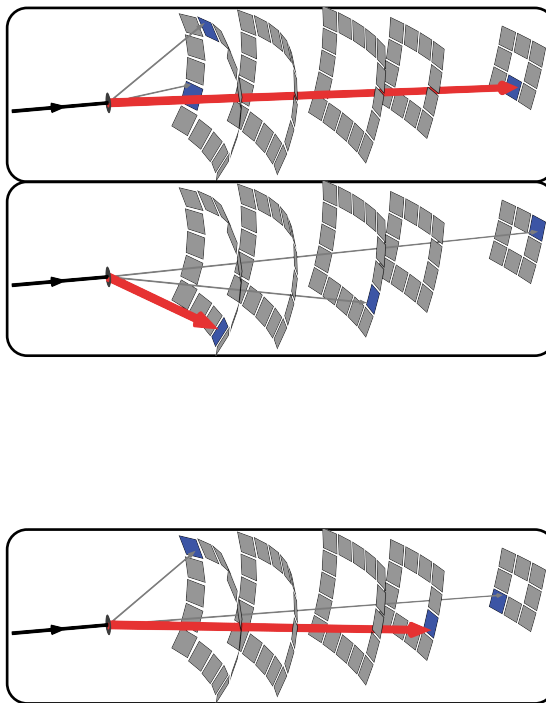
Creating a Mixed Event

Real Events

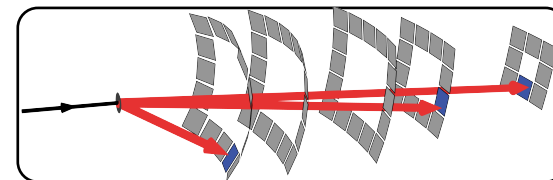


⋮

Selected Events



Mixed Events



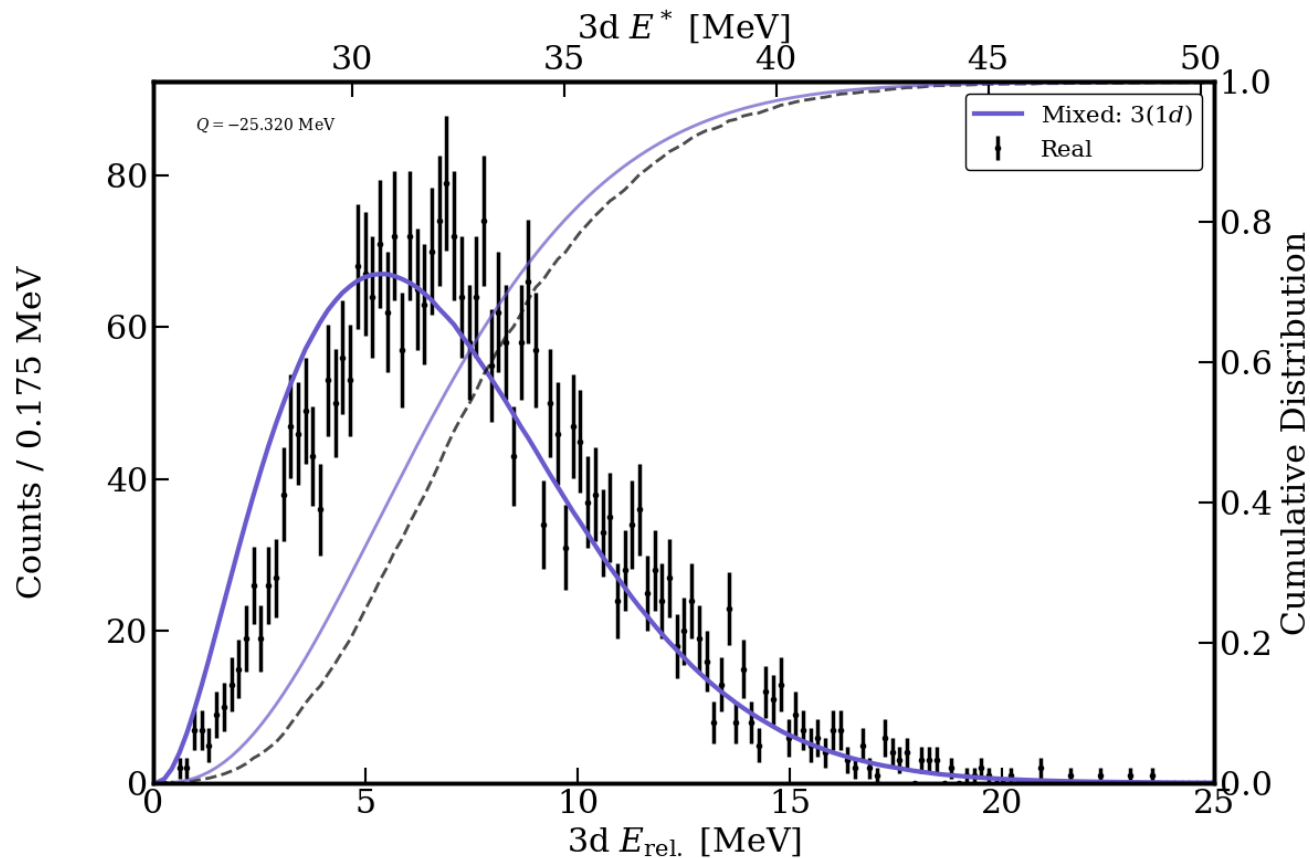
⋮

- Start w/ events containing all particles
- Select particles from separate events
- Construct new event

One way we know mixed events are imperfect (for this purpose)

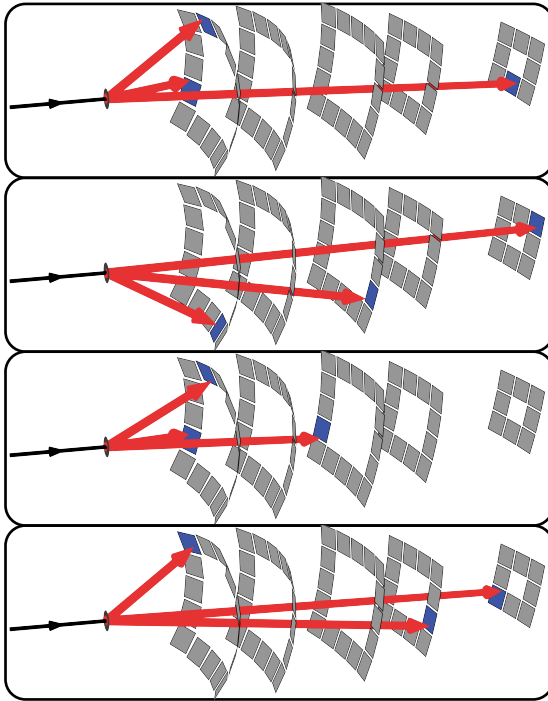
Mixed $\neq$ Total

- Mixing events removes particle-particle correlations, including:
 - 3-particle resonances (good)
 - Coulomb repulsion (bad)
 - 2-particle resonances (bad)
- Need a way to **remove 3-particle** correlations while **preserving 2-particle** correlations



Creating a *Partially Mixed Event*

Real Events

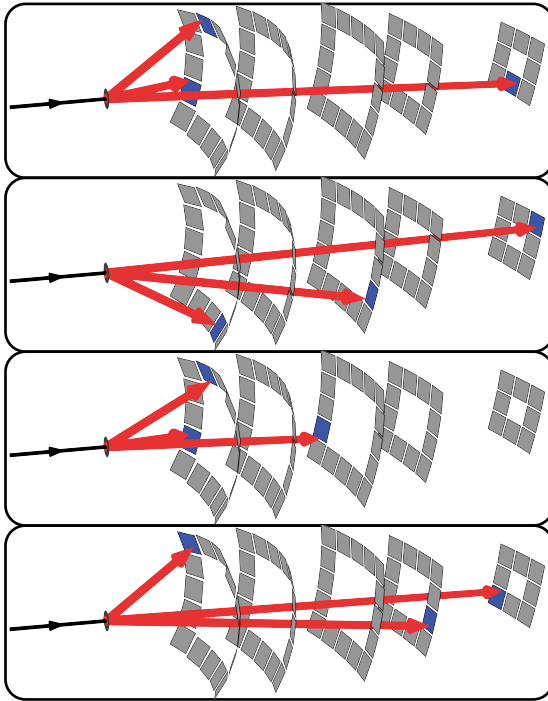


⋮

- Start w/ events containing all particles

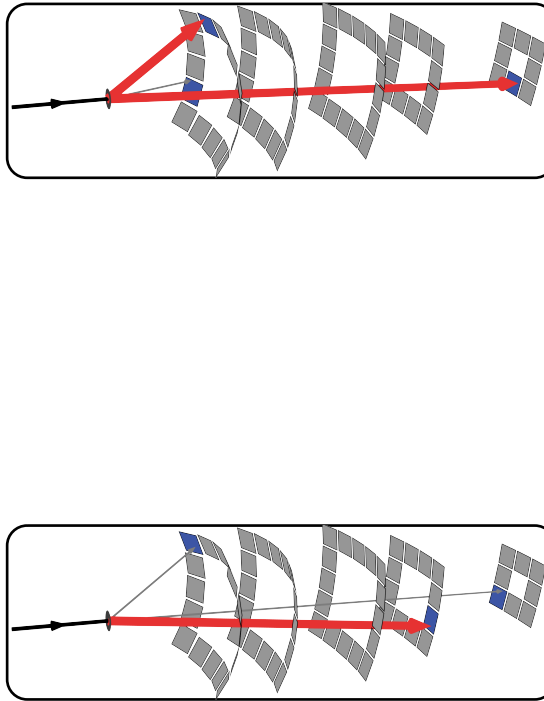
Creating a *Partially Mixed Event*

Real Events



⋮

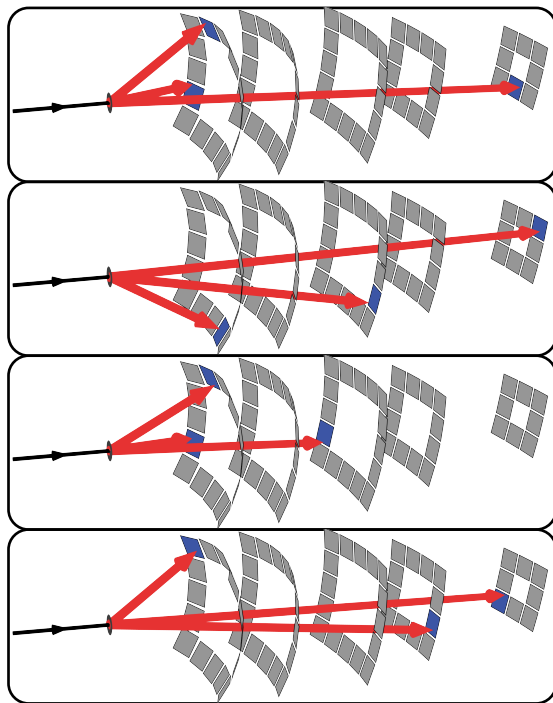
Selected Events



- Start w/ events containing all particles
- Select particles from separate events

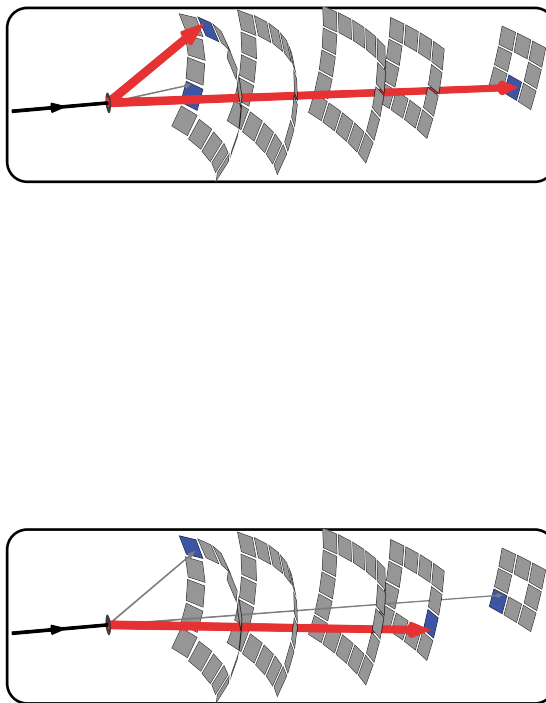
Creating a *Partially Mixed Event*

Real Events

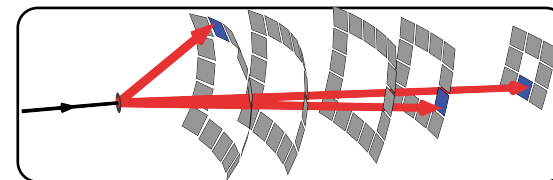


⋮

Selected Events



Partially Mixed Events

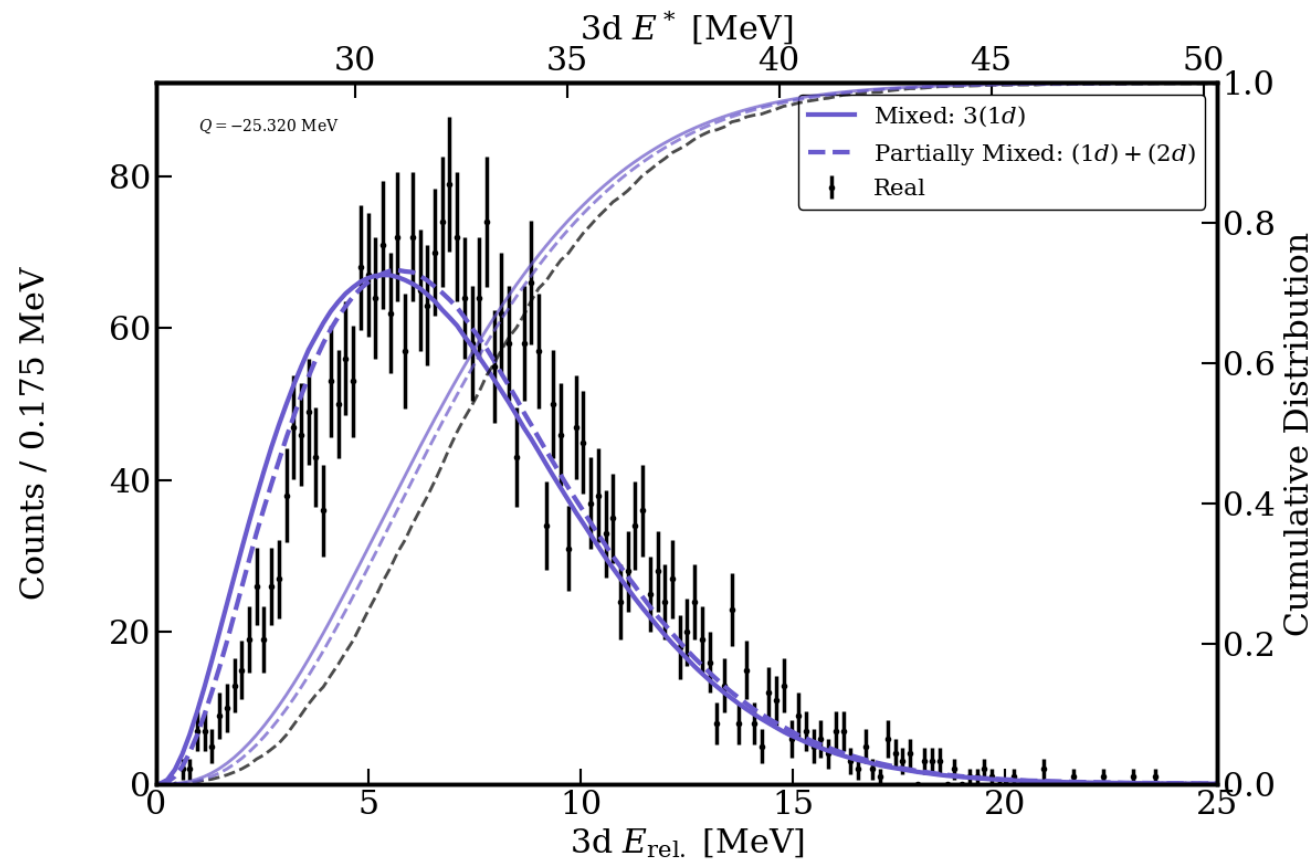


⋮

- Start w/ events containing all particles
- Select particles from separate events
- Construct new event

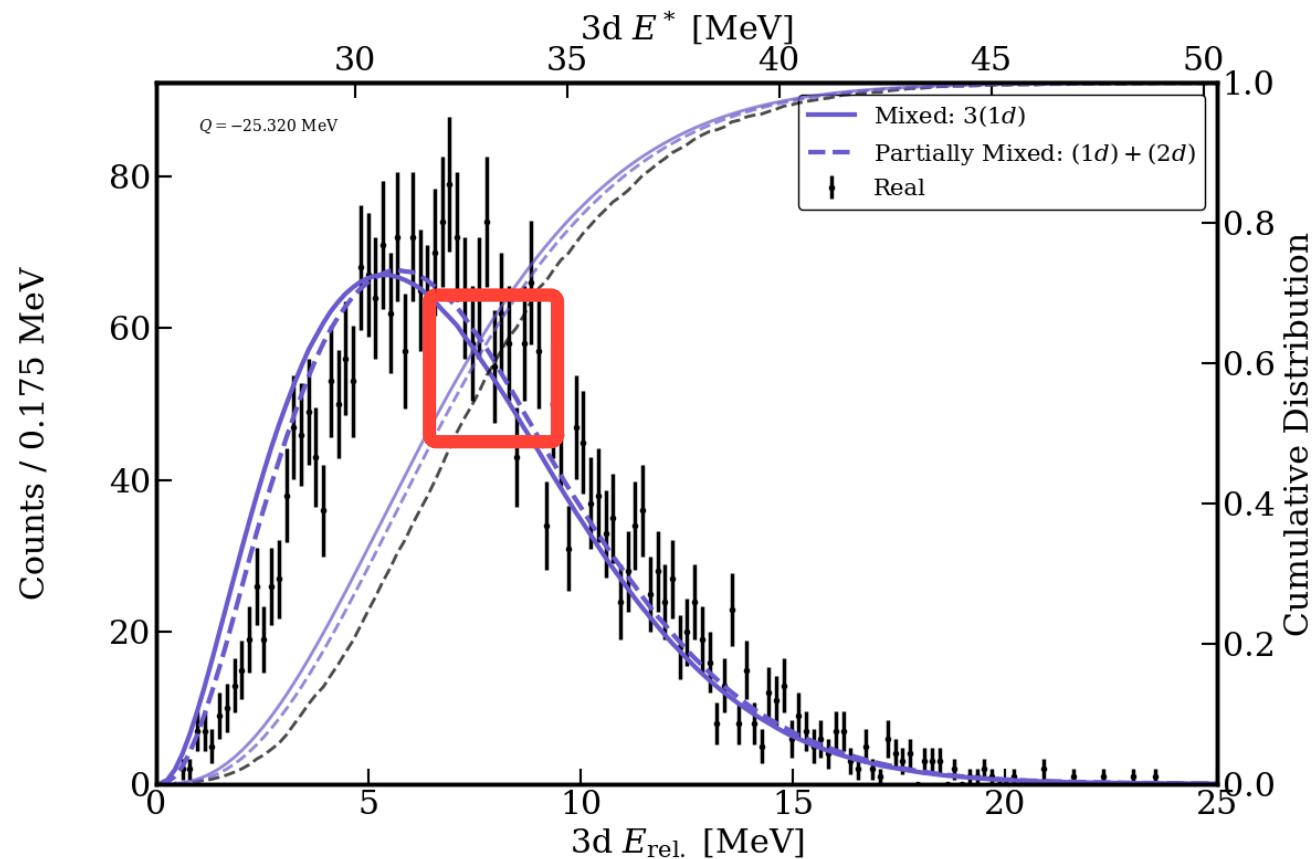
Incorporating one of the d - d correlations

- Partial mixing (PM) provides a minor improvement over full mixing (FM).



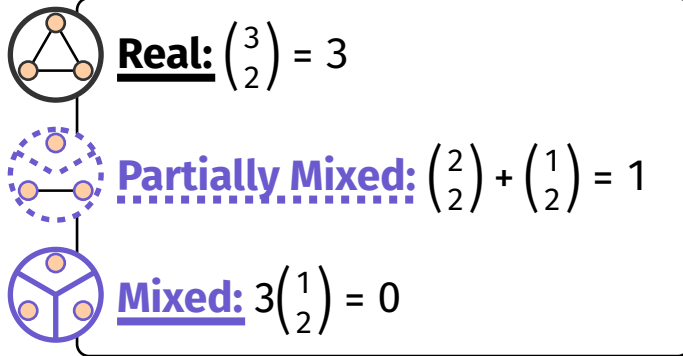
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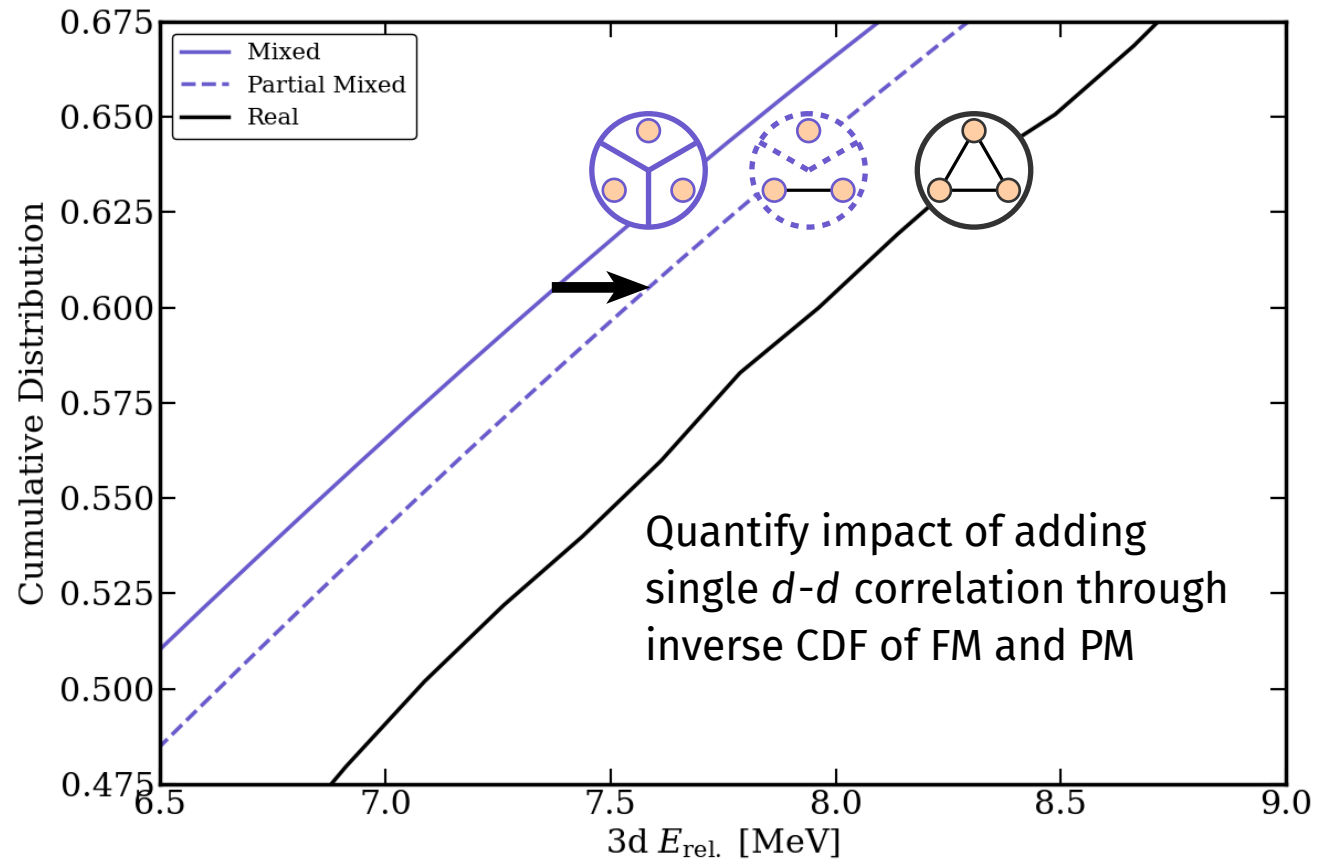


Measure the impact of one $d-d$ correlation

Number of $d-d$ correlations

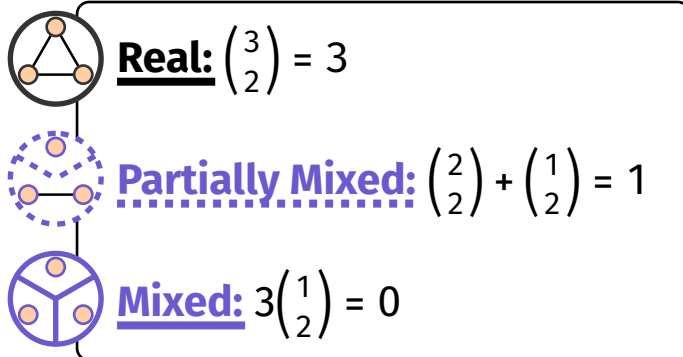


- Quantile mapping: Horizontal shifts in the CDF
 - Frequently used in climate change studies

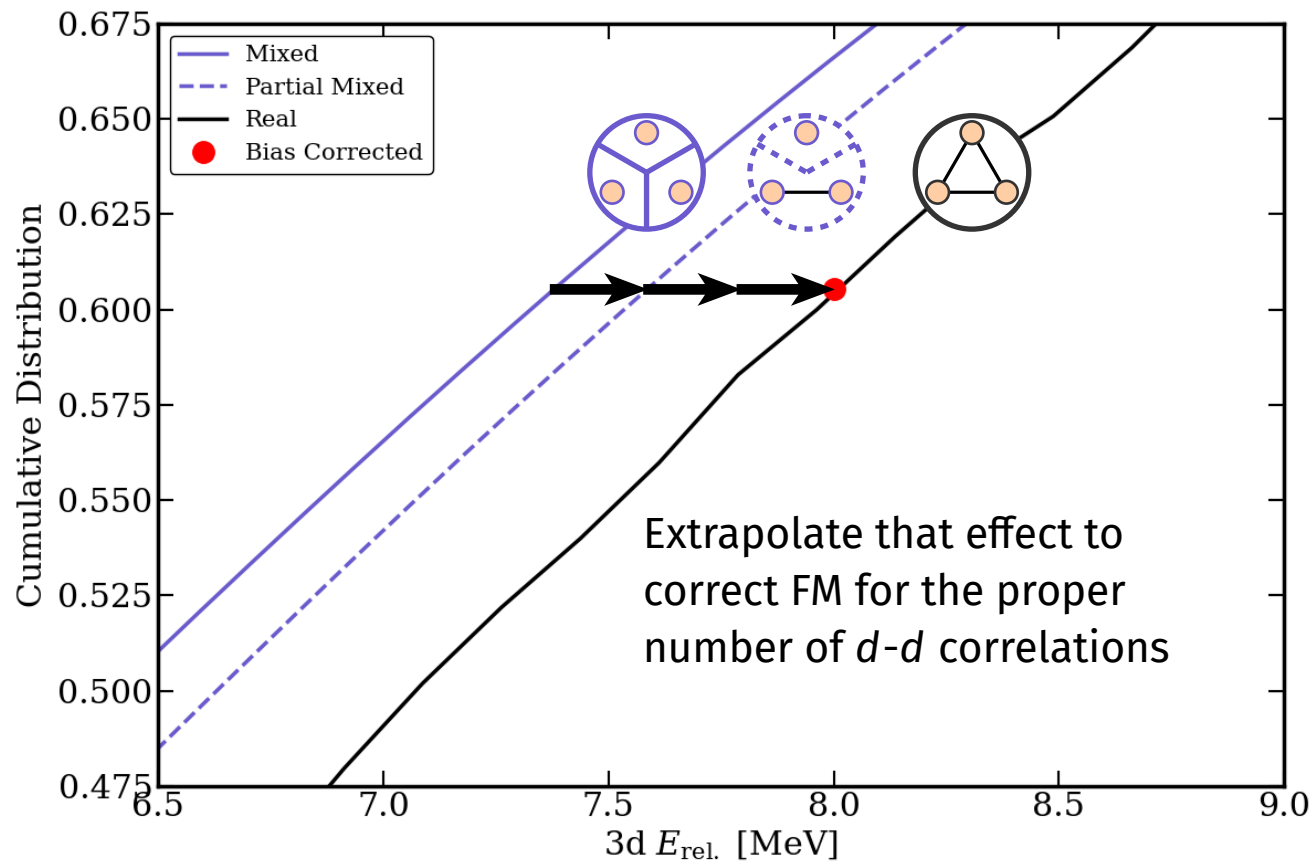


Extrapolate the impact to three $d-d$ correlations

Number of $d-d$ correlations

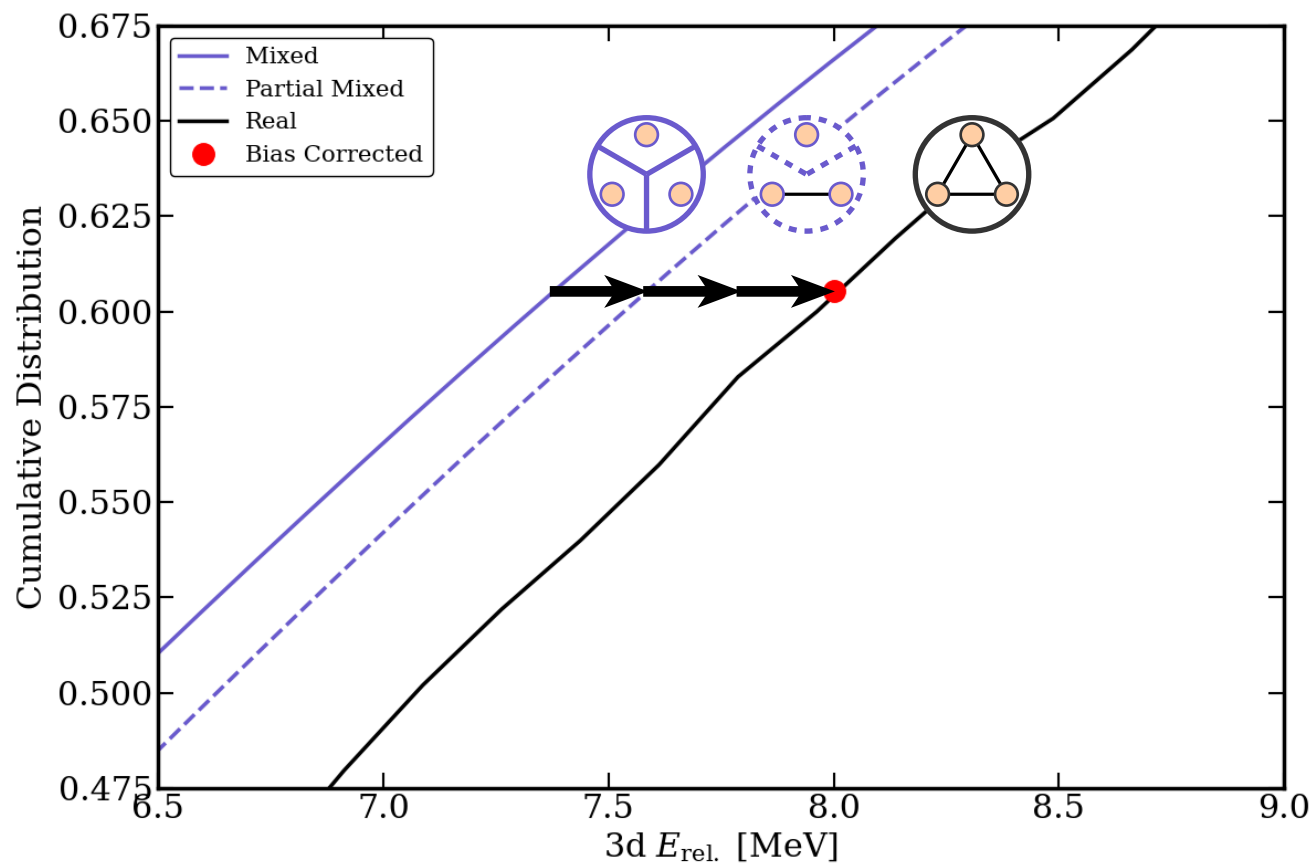


- Quantile mapping: Horizontal shifts in the CDF
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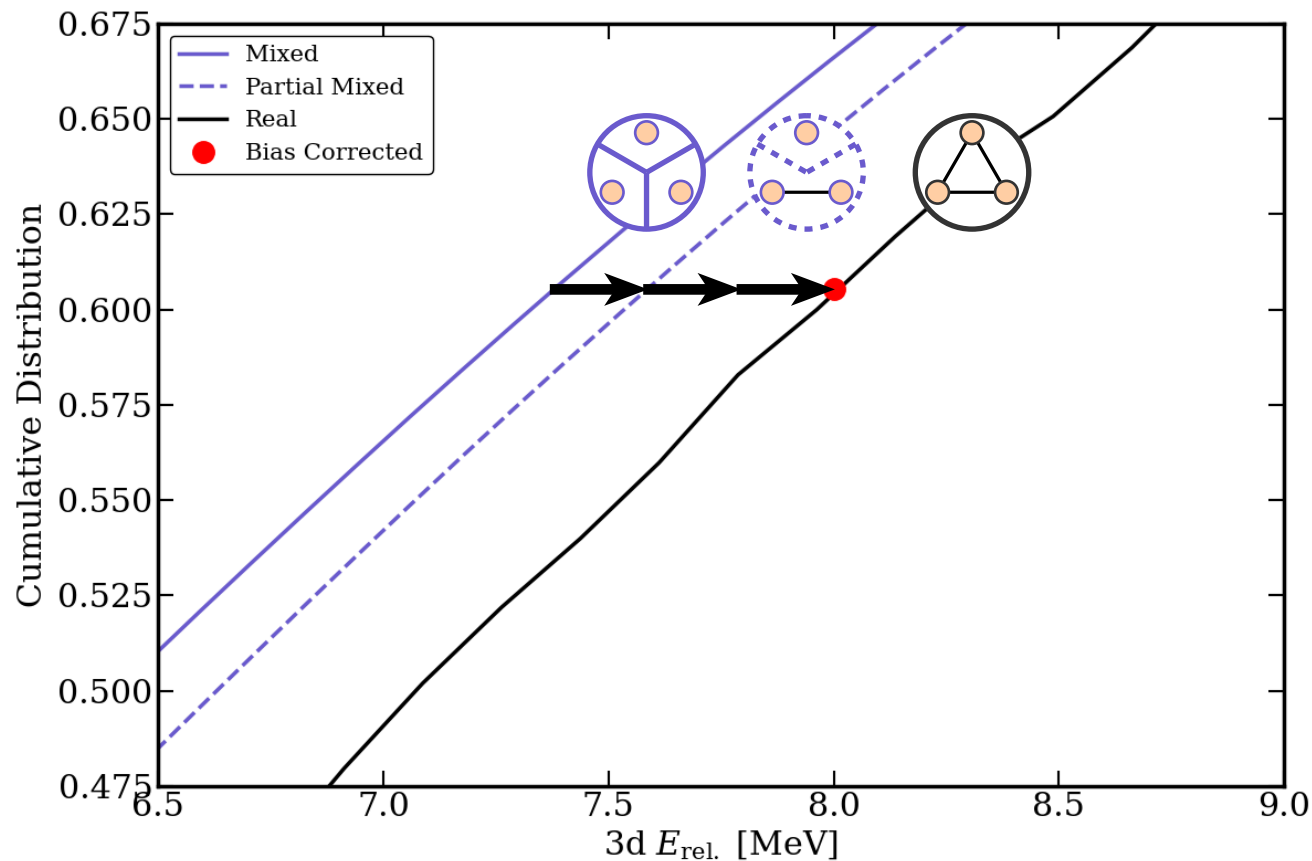
Extrapolate the impact to three d - d correlations

$B_{ddd}(E)$	Background estimate for ddd
$M_{\emptyset}(E)$	Fully Mixed Distribution
$M_{dd}(E)$	Partially Mixed Distribution



Extrapolate the impact to three d - d correlations

$B_{ddd}(E)$	Background estimate for ddd
$M_{\emptyset}(E)$	Fully Mixed Distribution
$M_{dd}(E)$	Partially Mixed Distribution
$\mathcal{B}_{ddd}^{-1}(q)$	Inverse CDF of $B_{ddd}(E)$
$\mathcal{M}_{\emptyset}^{-1}(q)$	Inverse CDF of $M_{\emptyset}(E)$
$\mathcal{M}_{dd}^{-1}(q)$	Inverse CDF of $M_{dd}(E)$



Extrapolate the impact to three d - d correlations

$B_{ddd}(E)$ Background estimate
for ddd

$M_{\emptyset}(E)$ Fully Mixed
Distribution

$M_{dd}(E)$ Partially Mixed
Distribution

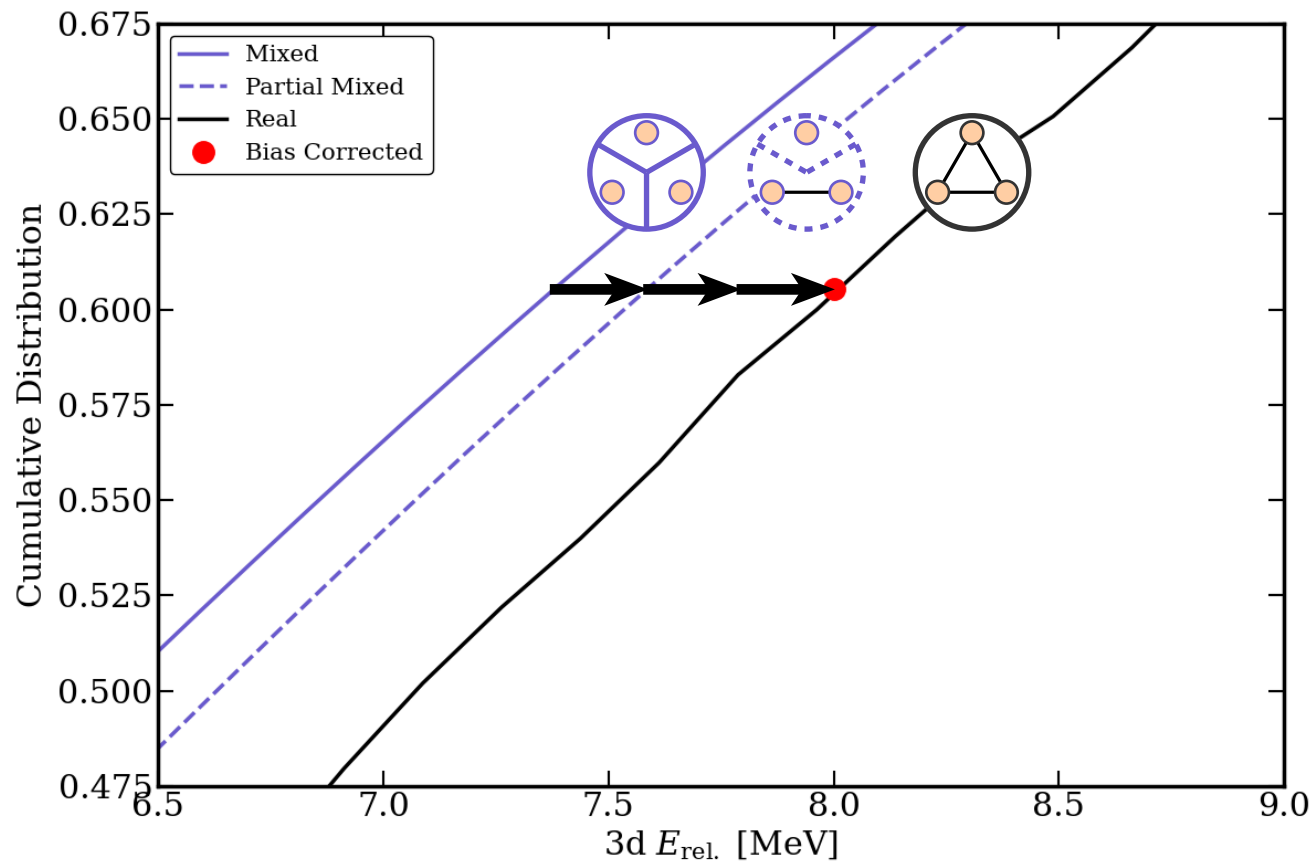
$\mathcal{B}_{ddd}^{-1}(q)$ Inverse CDF of $B_{ddd}(E)$

$\mathcal{M}_{\emptyset}^{-1}(q)$ Inverse CDF of $M_{\emptyset}(E)$

$\mathcal{M}_{dd}^{-1}(q)$ Inverse CDF of $M_{dd}(E)$

$$\mathcal{B}_{ddd}^{-1}(q) = \mathcal{M}_{\emptyset}^{-1}(q) + 3[\mathcal{M}_{dd}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)]$$

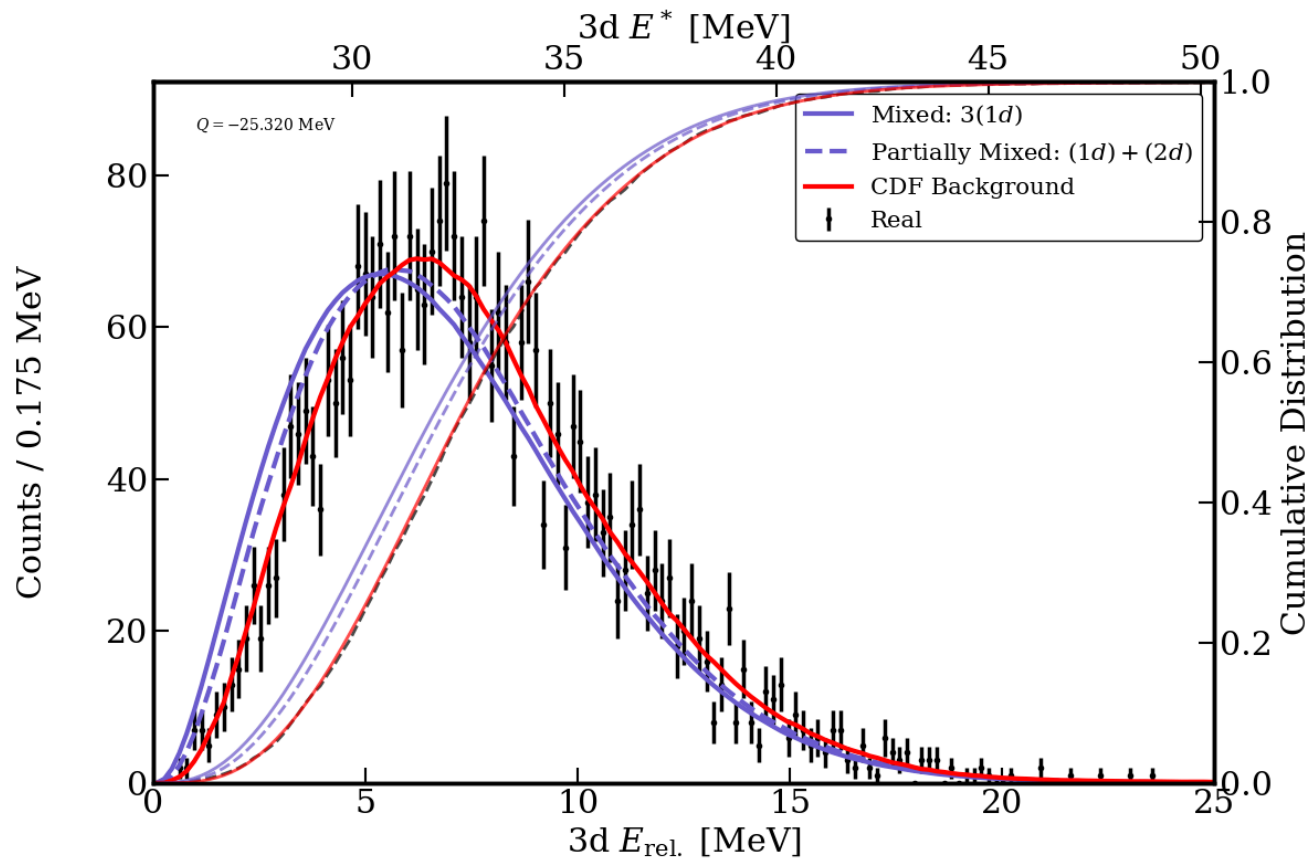
$$B_{ddd}(E_{\text{rel}}) \propto \frac{d}{dE_{\text{rel}}} \mathcal{B}_{ddd}(E_{\text{rel}})$$



Derive new background through CDF

Estimation of the Nonresonant Background

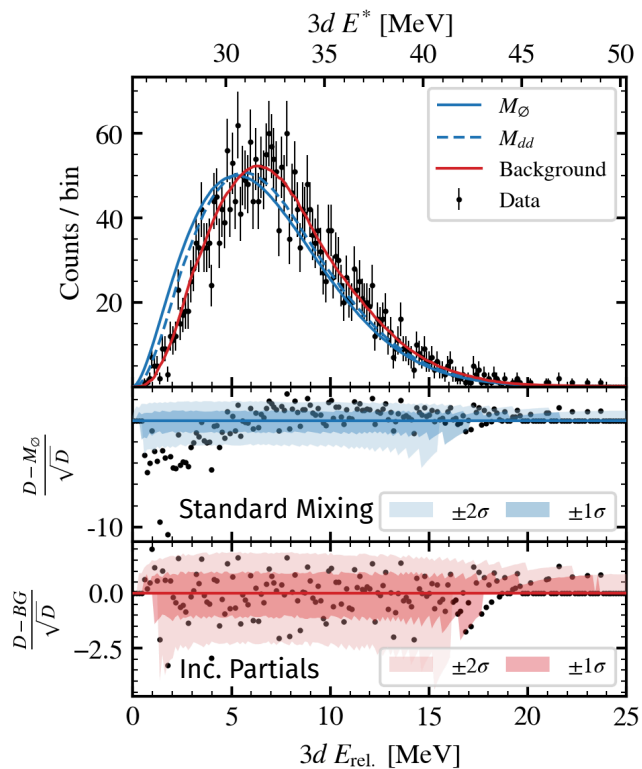
1. Measure the systematic impact of introducing a single 2-particle correlation via the transformation from FM to PM
2. Propagate that systematic change an additional 2 times to account for all three 2-particle correlations



Three 3d Correlations

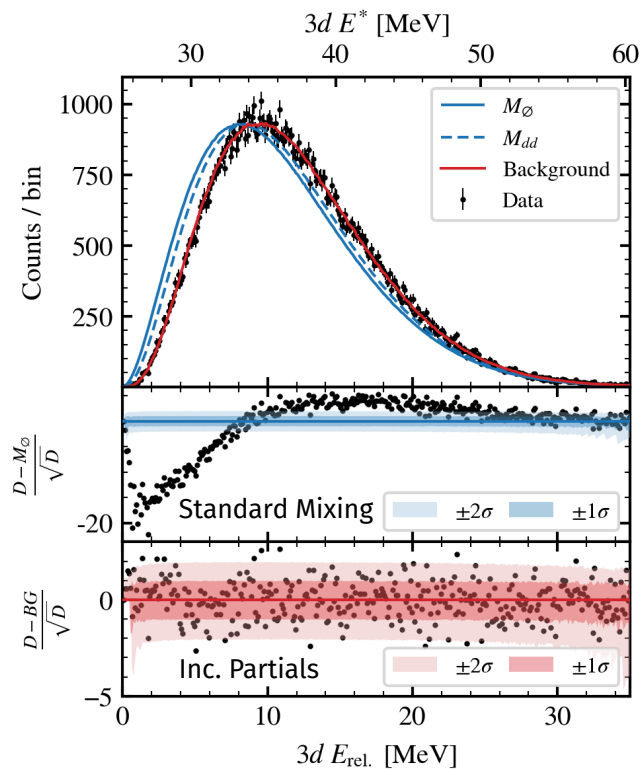
FAUST

$^{16}\text{O} + ^{12}\text{C} @ 35 \text{ MeV/u}$



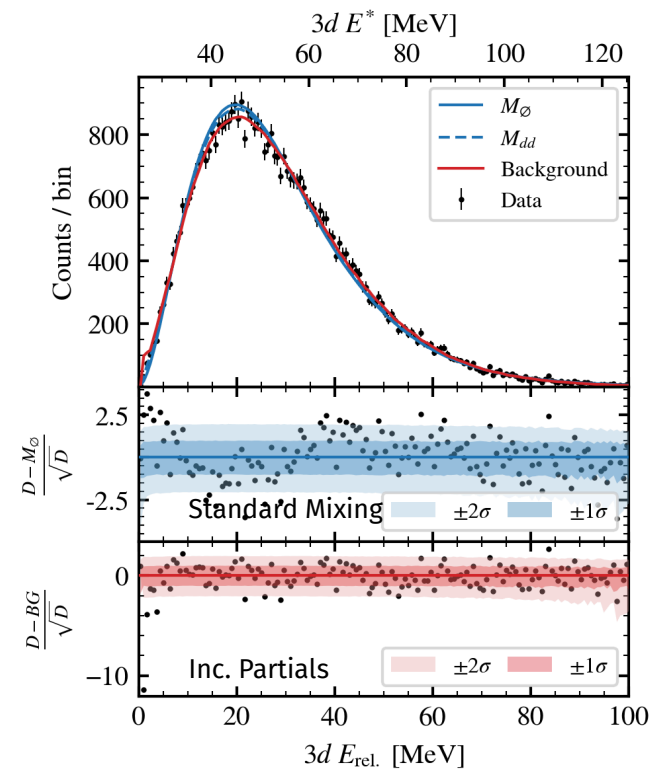
FAUST

$^{14}\text{N} + ^{10}\text{B} @ 28.3 \text{ MeV/u}$



NIMROD

$^{28}\text{Si} + ^{12}\text{C} @ 35 \text{ MeV/u}$



Generalization of the Method (So Far)

3 Identical Particles

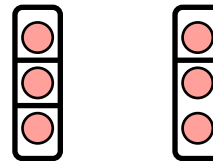
$$\mathcal{B}_{XXX}^{-1}(q) = \mathcal{M}_{\emptyset}^{-1}(q) + 3[\mathcal{M}_{XX}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)]$$



Generalization of the Method (So Far)

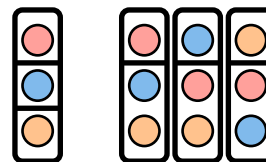
3 Identical Particles

$$\mathcal{B}_{XXX}^{-1}(q) = \mathcal{M}_{\emptyset}^{-1}(q) + 3[\mathcal{M}_{XX}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)]$$



3 Non-identical Particles

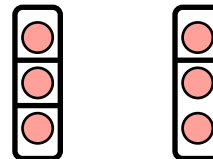
$$\begin{aligned}\mathcal{B}_{XYZ}^{-1}(q) = & \mathcal{M}_{\emptyset}^{-1}(q) + [\mathcal{M}_{XY}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)] \\ & + [\mathcal{M}_{YZ}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)] \\ & + [\mathcal{M}_{XZ}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)]\end{aligned}$$



Generalization of the Method (So Far)

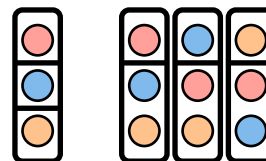
3 Identical Particles

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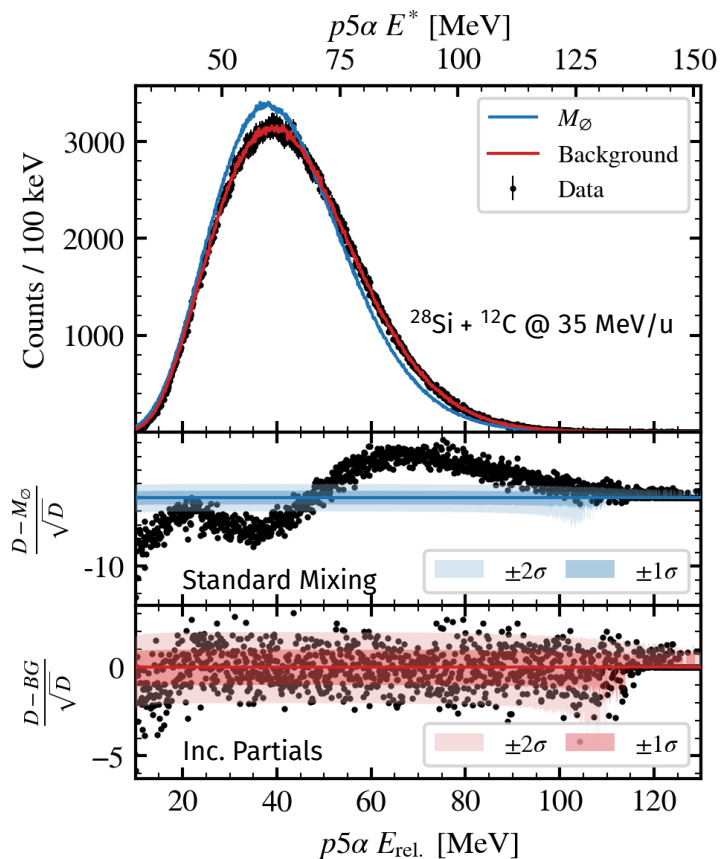


- P is a multiset of particles (e.g., $P = [d, d, d]$)
- $M_{P \setminus \{i\}}(E)$ is the energy distribution for mixed events where particle i was swapped.

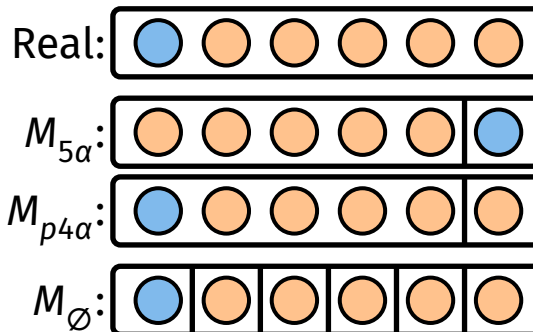
3 or More Generic Particles

$$\mathcal{B}_P^{-1}(q) = \mathcal{M}_{\emptyset}^{-1}(q) + \frac{1}{|P| - 2} \sum_{i \in P} [\mathcal{M}_{P \setminus \{i\}}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)]$$

Example with $p5\alpha - P = [p, \alpha, \alpha, \alpha, \alpha, \alpha]$



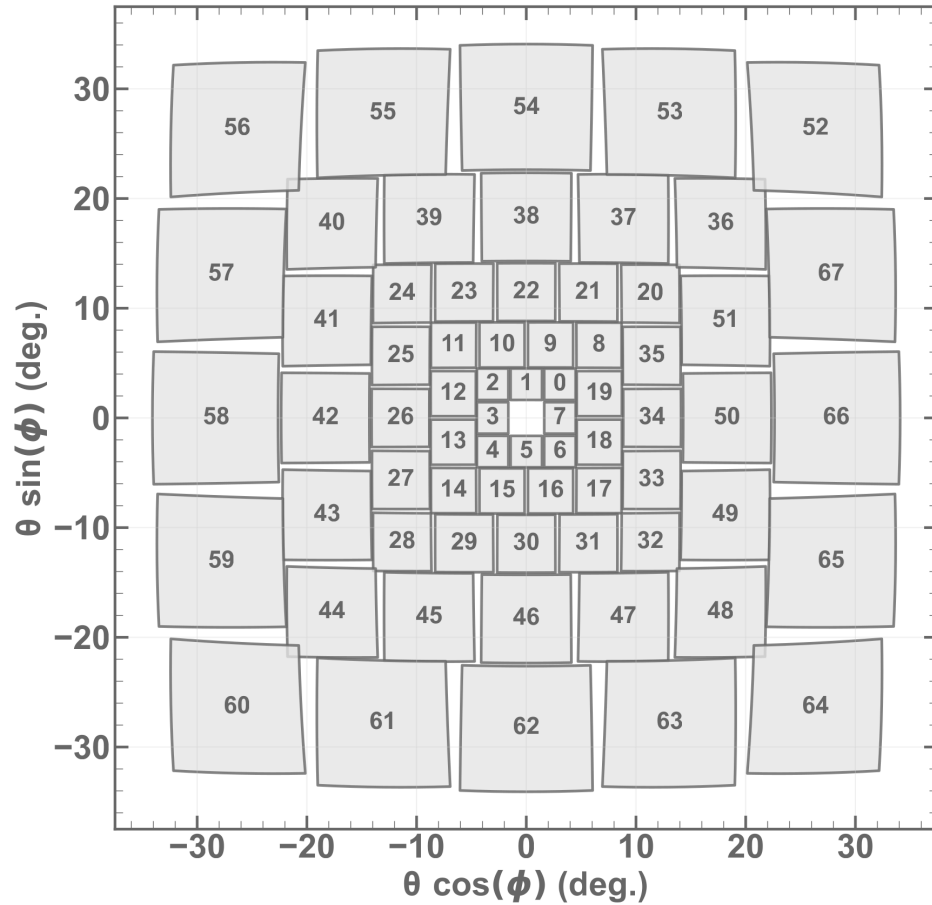
$$\mathcal{B}_P^{-1}(q) = \mathcal{M}_{\emptyset}^{-1}(q) + \frac{1}{|P| - 2} \sum_{i \in P} [\mathcal{M}_{P \setminus \{i\}}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)]$$



$$\begin{aligned} \mathcal{B}_{p5\alpha}^{-1}(q) &= \mathcal{M}_{\emptyset}^{-1}(q) \\ &+ \frac{1}{4} [\mathcal{M}_{5\alpha}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)] \\ &+ \frac{5}{4} [\mathcal{M}_{p4\alpha}^{-1}(q) - \mathcal{M}_{\emptyset}^{-1}(q)] \end{aligned}$$

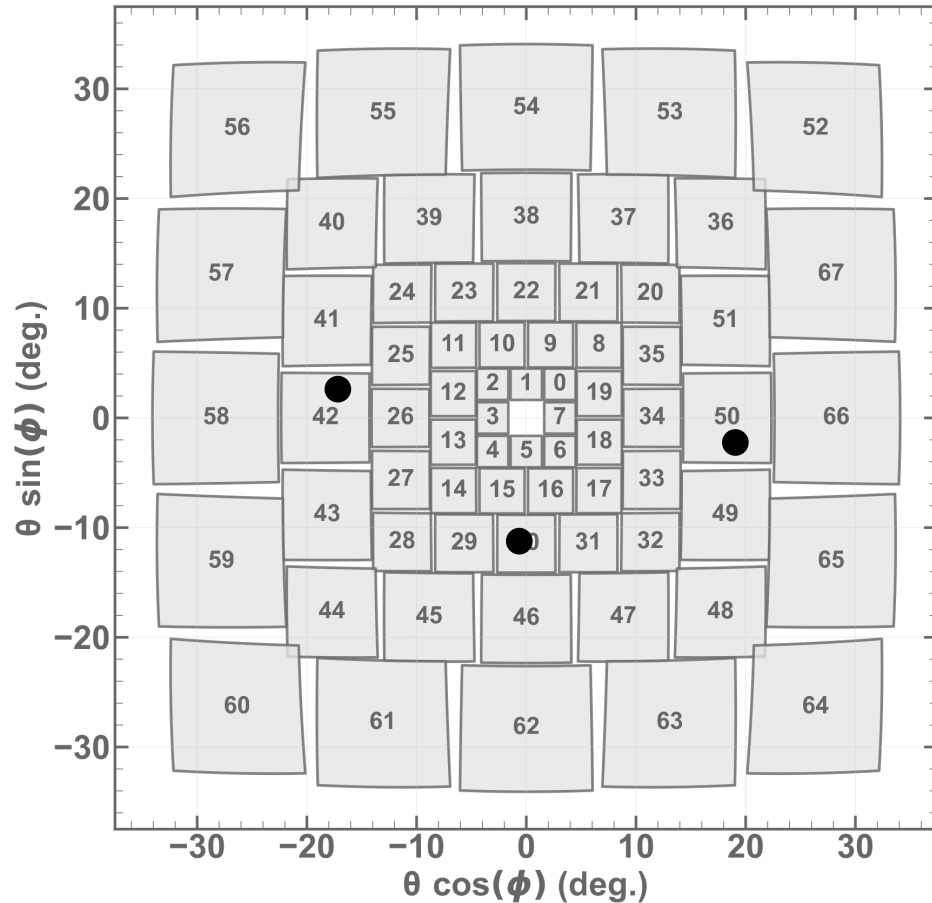
- Standard mixing fails to describe the distribution
- Current work gets **much** closer, but still slightly off at low energy

Another Problem with Traditional Mixing



- Consider 3 identical particles again
- Standard Partial Mixing Procedure

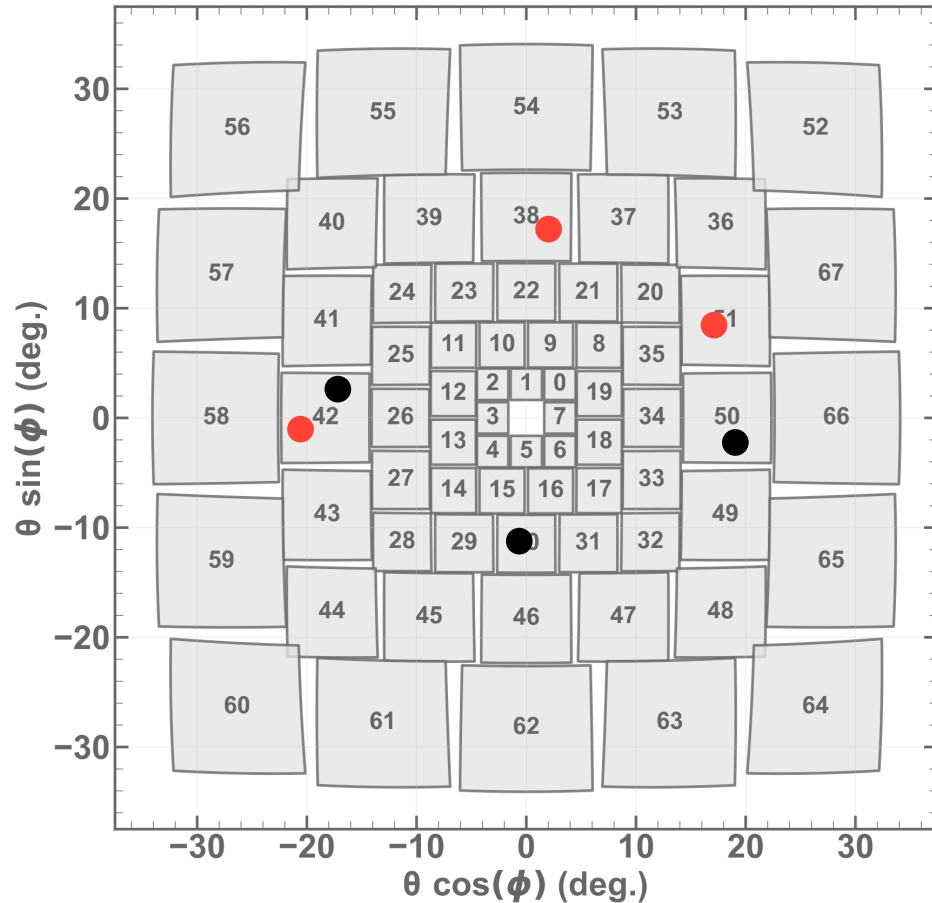
Another Problem with Traditional Mixing



Consider 3 identical particles again

- Standard Partial Mixing Procedure
 - Select one **event**

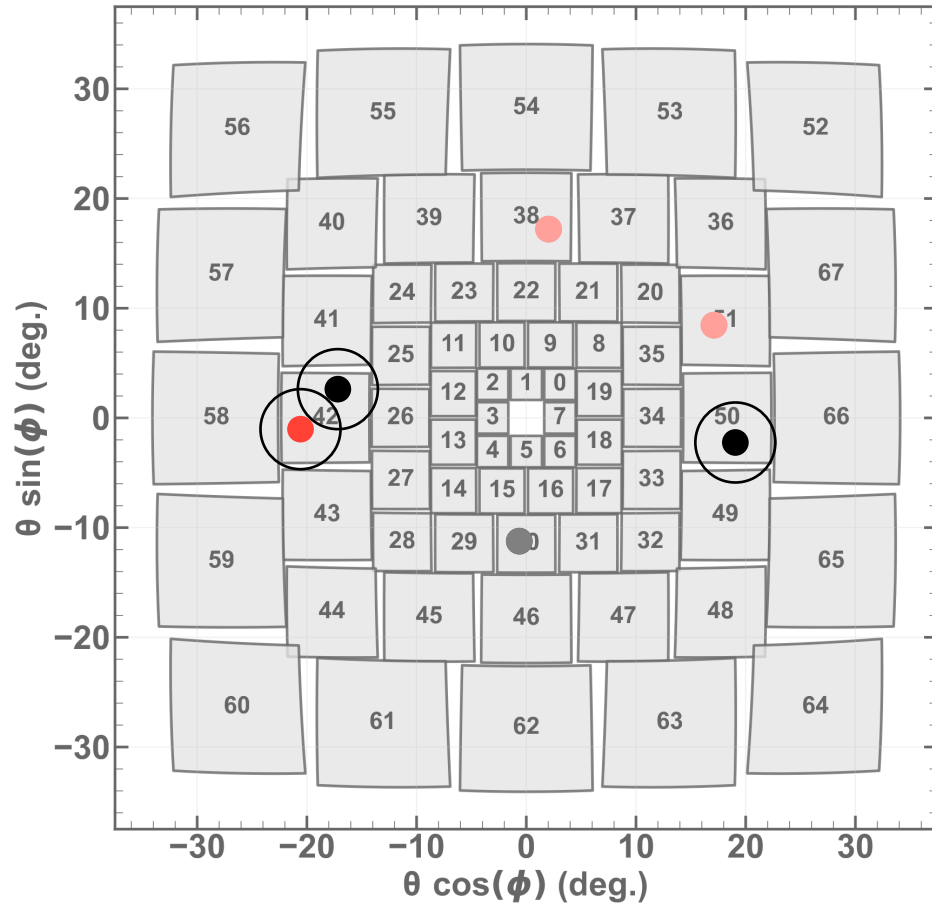
Another Problem with Traditional Mixing



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 - Select one **event**
 - Select another **event**

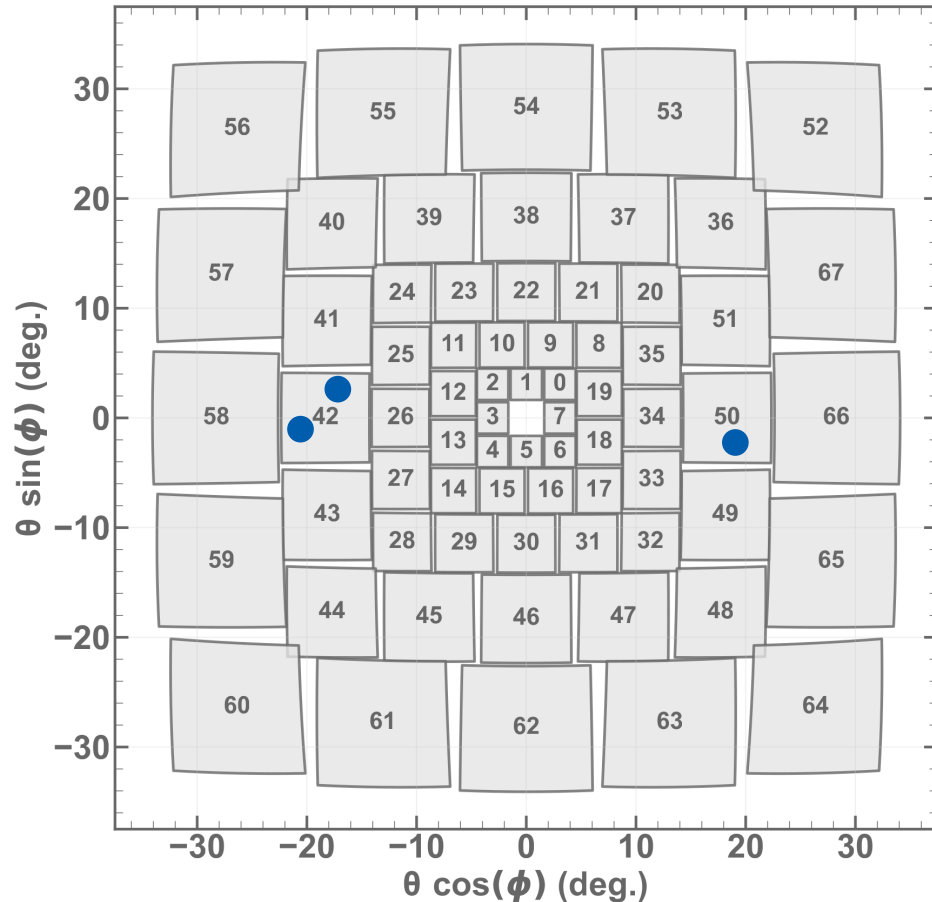
Another Problem with Traditional Mixing



Consider 3 identical particles again

- Standard Partial Mixing Procedure
 - Select one **event**
 - Select another **event**
 - Choose all but one particle from one **event** and the last particle from **the other event**

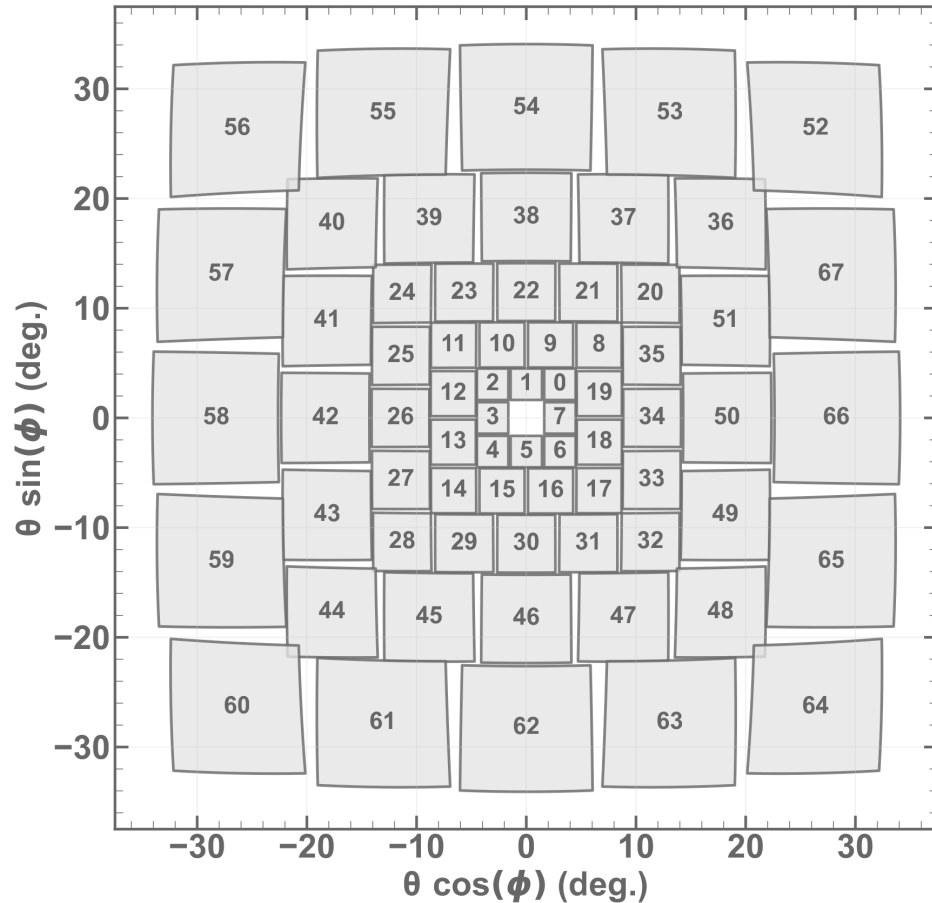
Another Problem with Traditional Mixing



Consider 3 identical particles again

- Standard Partial Mixing Procedure
 - Select one **event**
 - Select another **event**
 - Choose all but one particle from one **event** and the last particle from **the other event**
 - New **event** has a double hit!

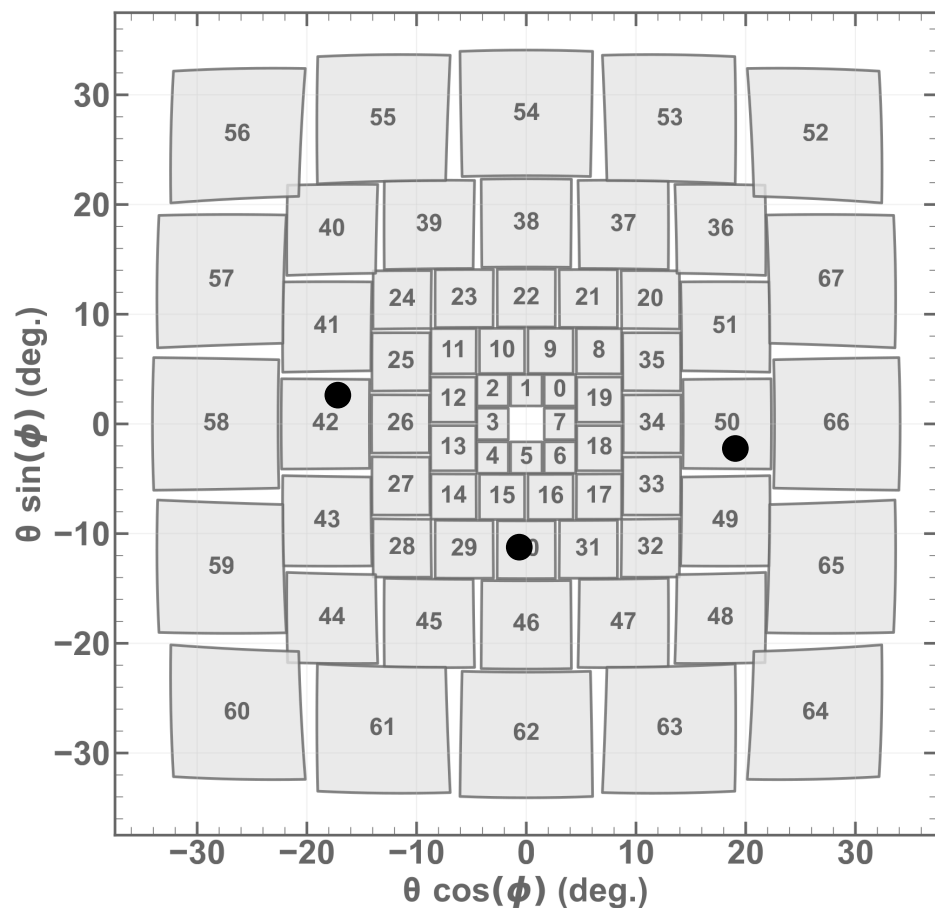
Hit Pattern Preservation (Alternate Mixing Technique)



Consider 3 identical particles again

- **Hit Pattern Preserving** Partial Mixing Procedure

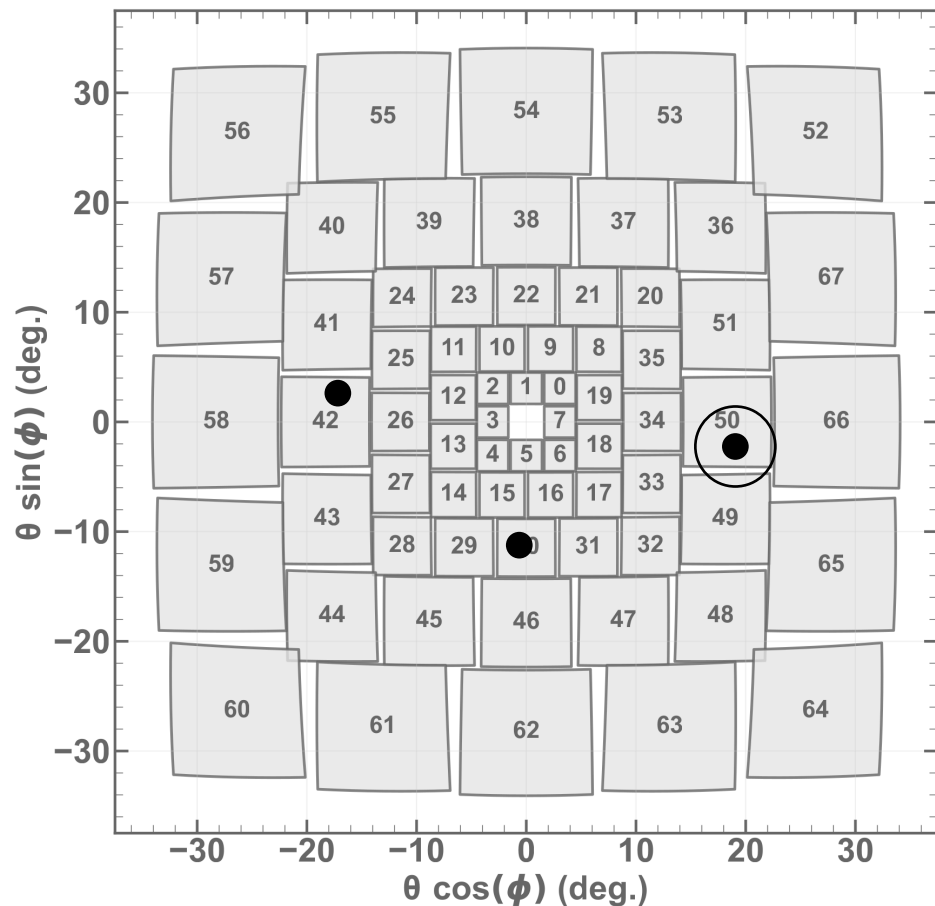
Hit Pattern Preservation (Alternate Mixing Technique)



Consider 3 identical particles again

- **Hit Pattern Preserving** Partial Mixing Procedure
 - Select one **event**

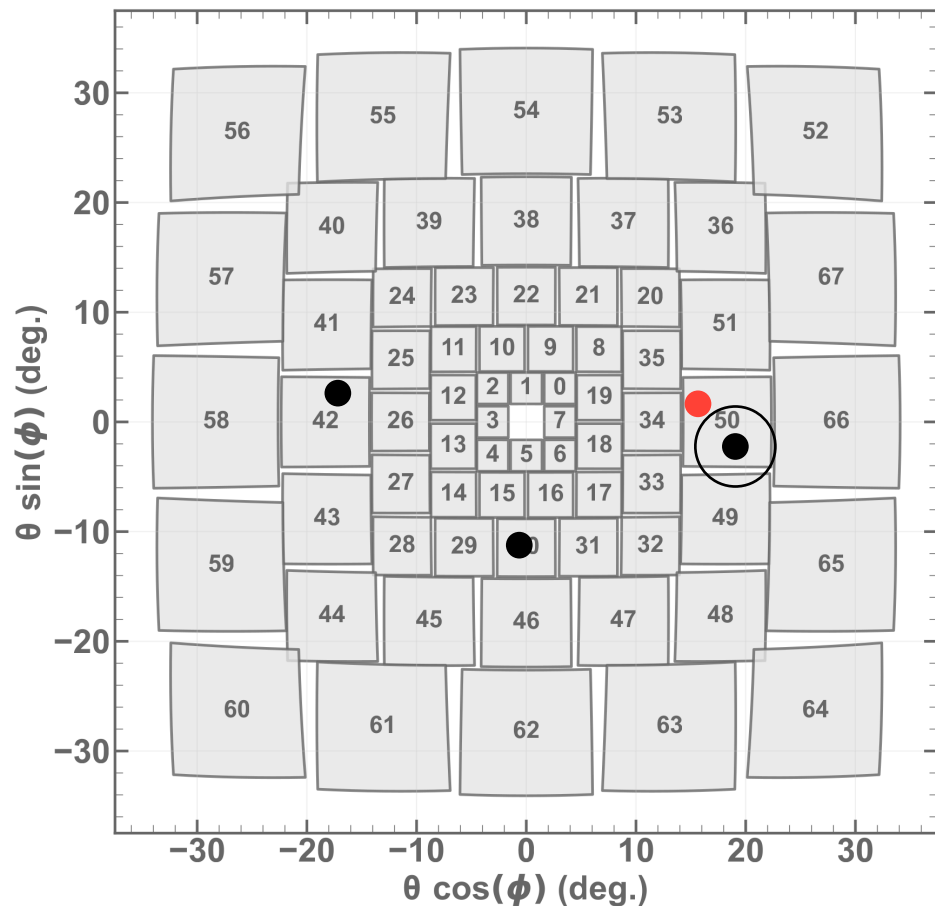
Hit Pattern Preservation (Alternate Mixing Technique)



Consider 3 identical particles again

- **Hit Pattern Preserving** Partial Mixing Procedure
 - Select one **event**
 - Identify particle that is *not* to be preserved

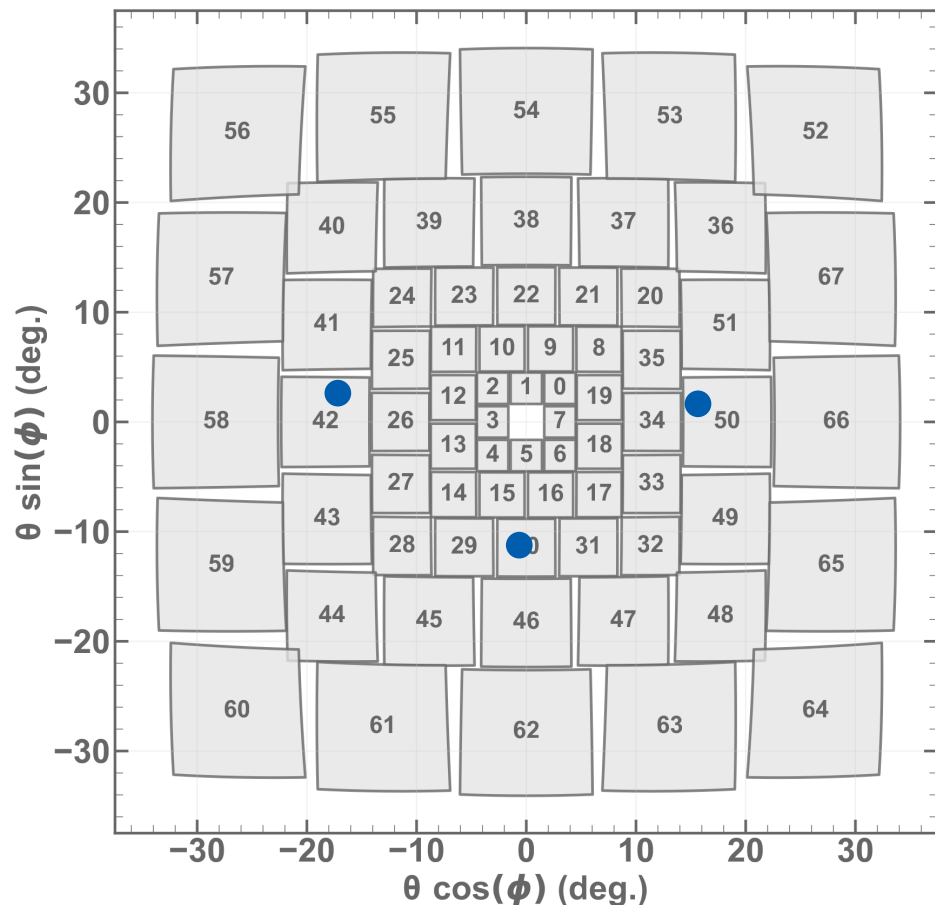
Hit Pattern Preservation (Alternate Mixing Technique)



Consider 3 identical particles again

- **Hit Pattern Preserving** Partial Mixing Procedure
 - Select one **event**
 - Identify particle that is *not* to be preserved
 - Find a particle of same type that was measured in another **event** by the *same detector*

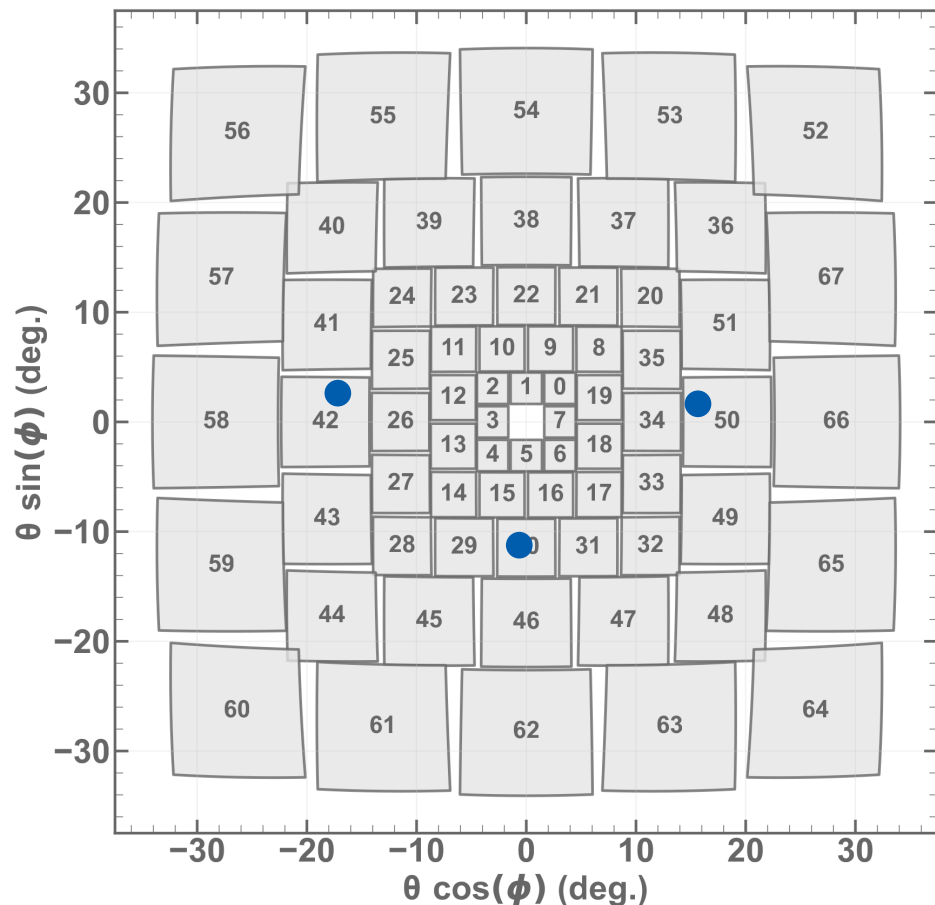
Hit Pattern Preservation (Alternate Mixing Technique)



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 - Find a particle of same type that was measured in another **event** by the *same detector*
 - New **event** has no double hits!
 - Has exact same hit pattern as real
 - For Full Mixing:
 - Swap each particle individually from all different events

Hit Pattern Preservation (Alternate Mixing Technique)

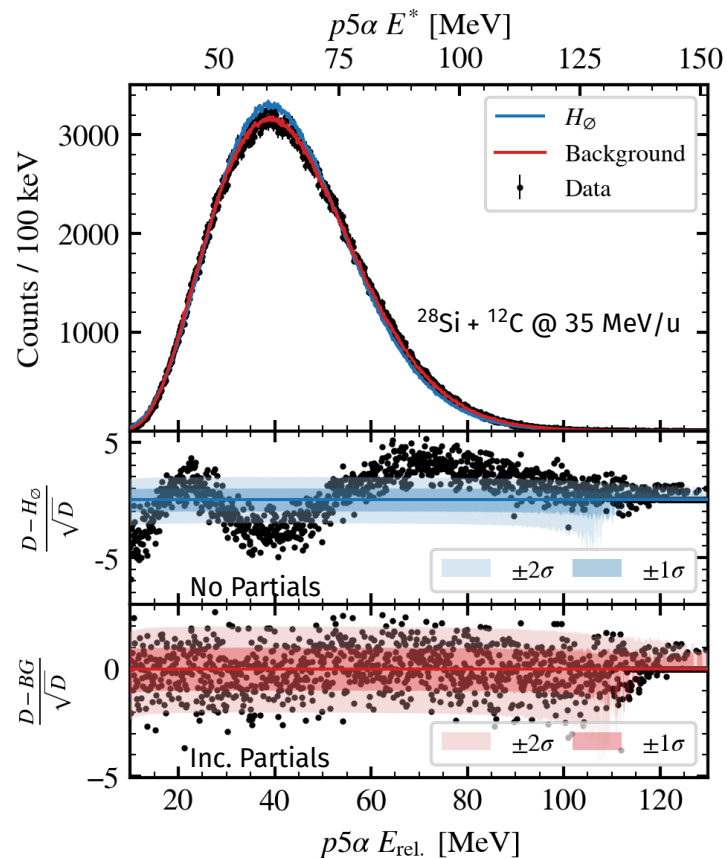


Consider 3 identical particles again

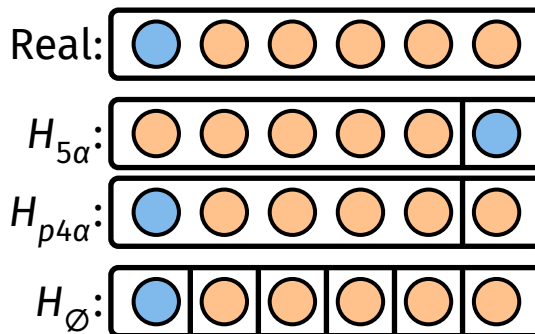
- **Hit Pattern Preserving** Partial Mixing Procedure
 - ▶ Select one **event**
 - ▶ Identify particle that is *not* to be preserved
 - ▶ Find a particle of same type that was measured in another **event** by the *same detector*
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$$\mathcal{B}_P^{-1}(q) = \mathcal{H}_{\emptyset}^{-1}(q) + \sum_{i \in P} w_i [\mathcal{H}_{P \setminus \{i\}}^{-1}(q) - \mathcal{H}_{\emptyset}^{-1}(q)]$$

Example with $p5\alpha - P = [p, \alpha, \alpha, \alpha, \alpha, \alpha]$



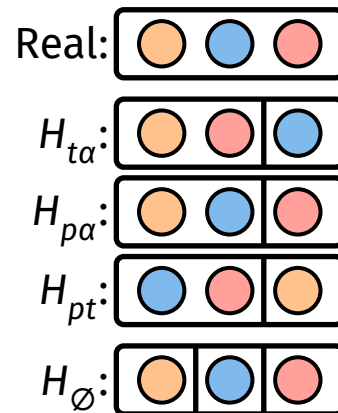
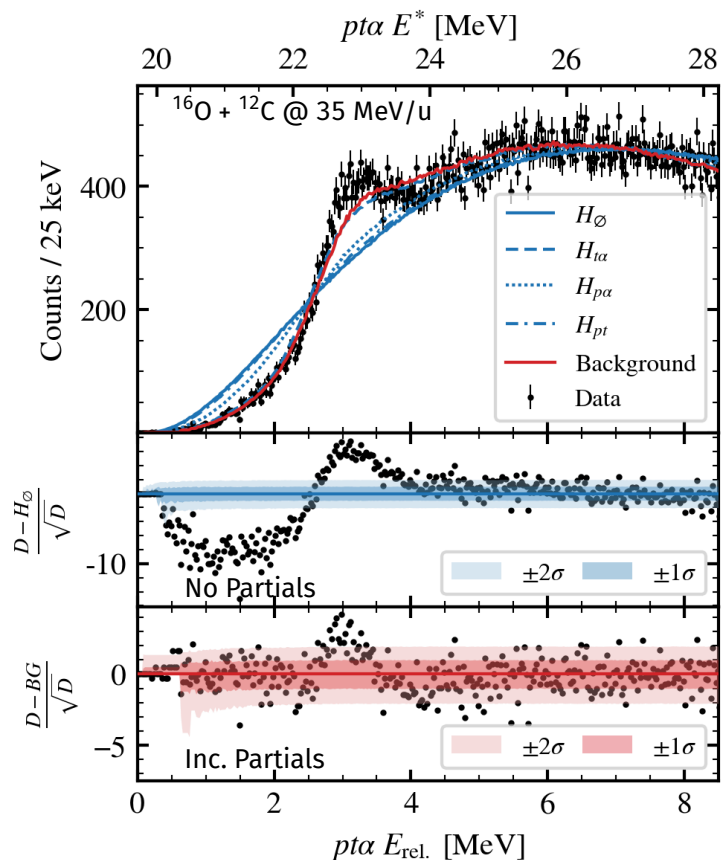
$$\mathcal{B}_P^{-1}(q) = \mathcal{H}_{\emptyset}^{-1}(q) + \sum_{i \in P} w_i [\mathcal{H}_{P \setminus \{i\}}^{-1}(q) - \mathcal{H}_{\emptyset}^{-1}(q)]$$



$$\begin{aligned} \mathcal{B}_{p5\alpha}^{-1}(q) &= \mathcal{H}_{\emptyset}^{-1}(q) \\ &+ 0.248 [\mathcal{H}_{5\alpha}^{-1}(q) - \mathcal{H}_{\emptyset}^{-1}(q)] \\ &+ 1.045 [\mathcal{H}_{p4\alpha}^{-1}(q) - \mathcal{H}_{\emptyset}^{-1}(q)] \end{aligned}$$

- $H_{\emptyset}$ is not enough to describe the background.
- $H_{\emptyset}$, $H_{p4\alpha}$, and $H_{5\alpha}$ in tandem are enough to describe the distribution very well.

Example with Populated Resonances: ${}^8\text{Be}^* \rightarrow p\alpha$



$$\mathcal{B}_{pt\alpha}^{-1}(q) = \mathcal{H}_{\emptyset}^{-1}(q)$$

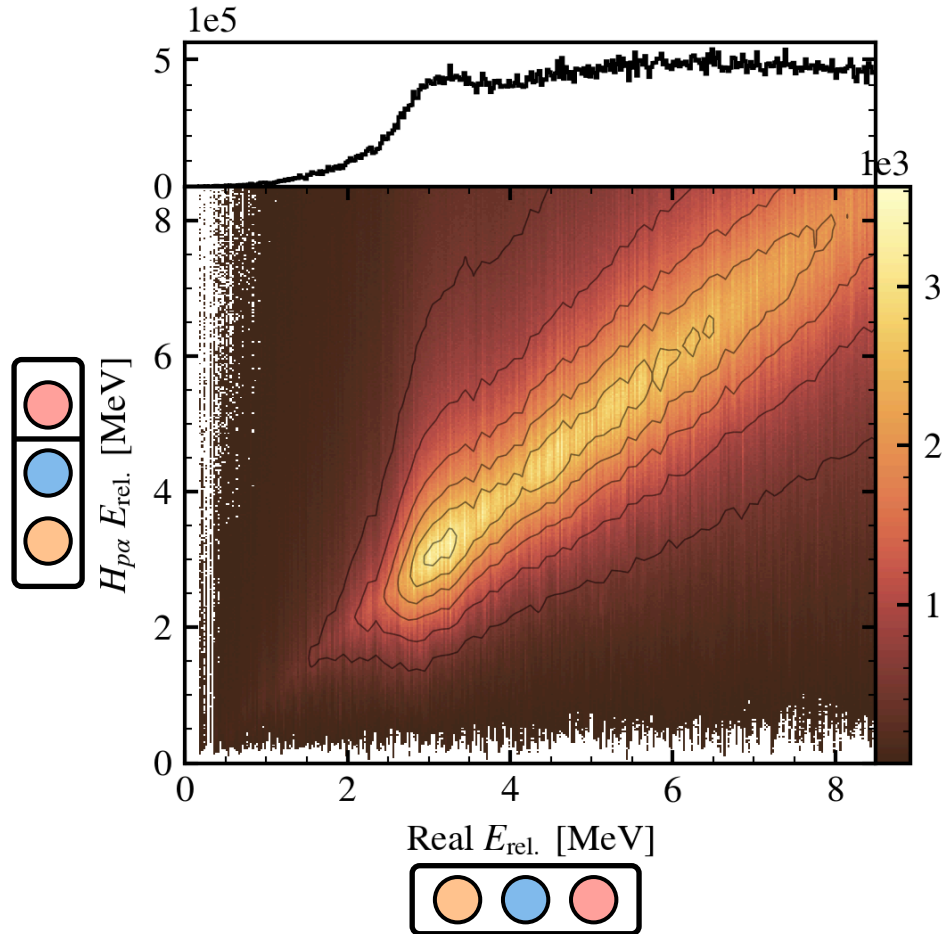
$$+w_p[\mathcal{H}_{t\alpha}^{-1}(q) - \mathcal{H}_{\emptyset}^{-1}(q)]$$

$$+w_t[\mathcal{H}_{p\alpha}^{-1}(q) - \mathcal{H}_{\emptyset}^{-1}(q)]$$

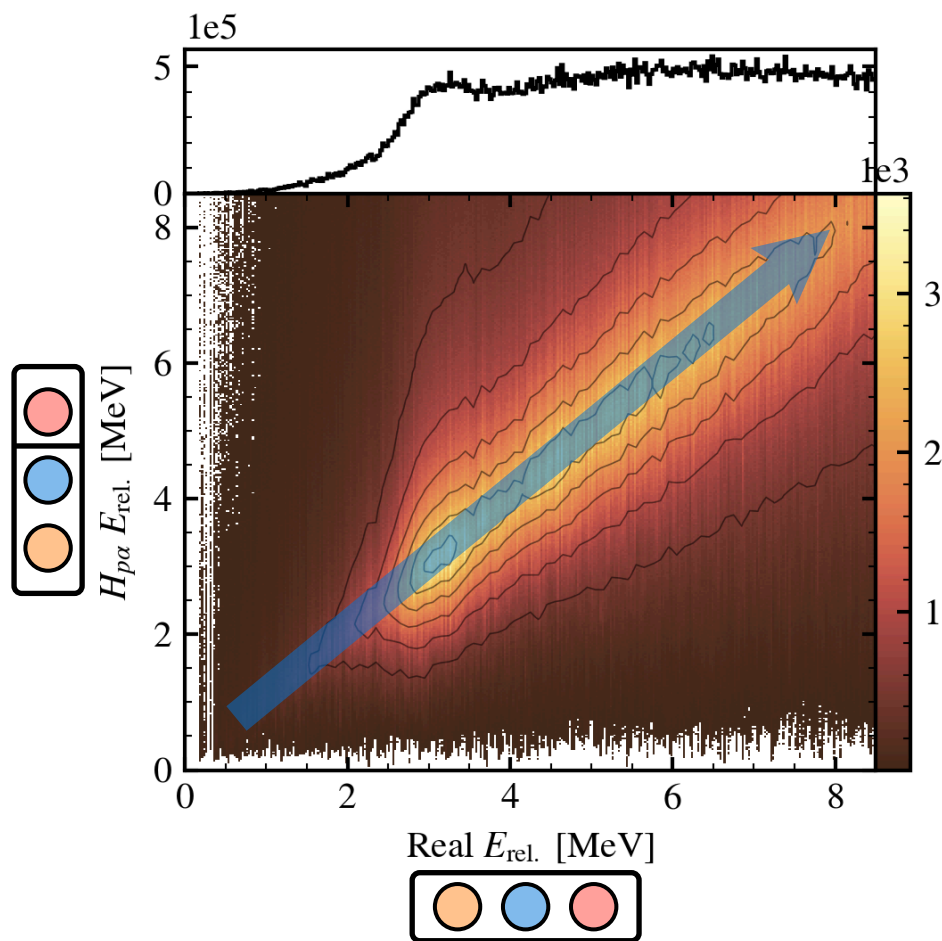
$$+w_{\alpha}[\mathcal{H}_{pt}^{-1}(q) - \mathcal{H}_{\emptyset}^{-1}(q)]$$

- $H_{\emptyset}$, $H_{p\alpha}$, $H_{t\alpha}$, and H_{pt} in tandem are **NOT** enough to describe the distribution very well.
 - Resonant contribution needs to be included

Impact of Real 3-particle Correlations on Mixed Events



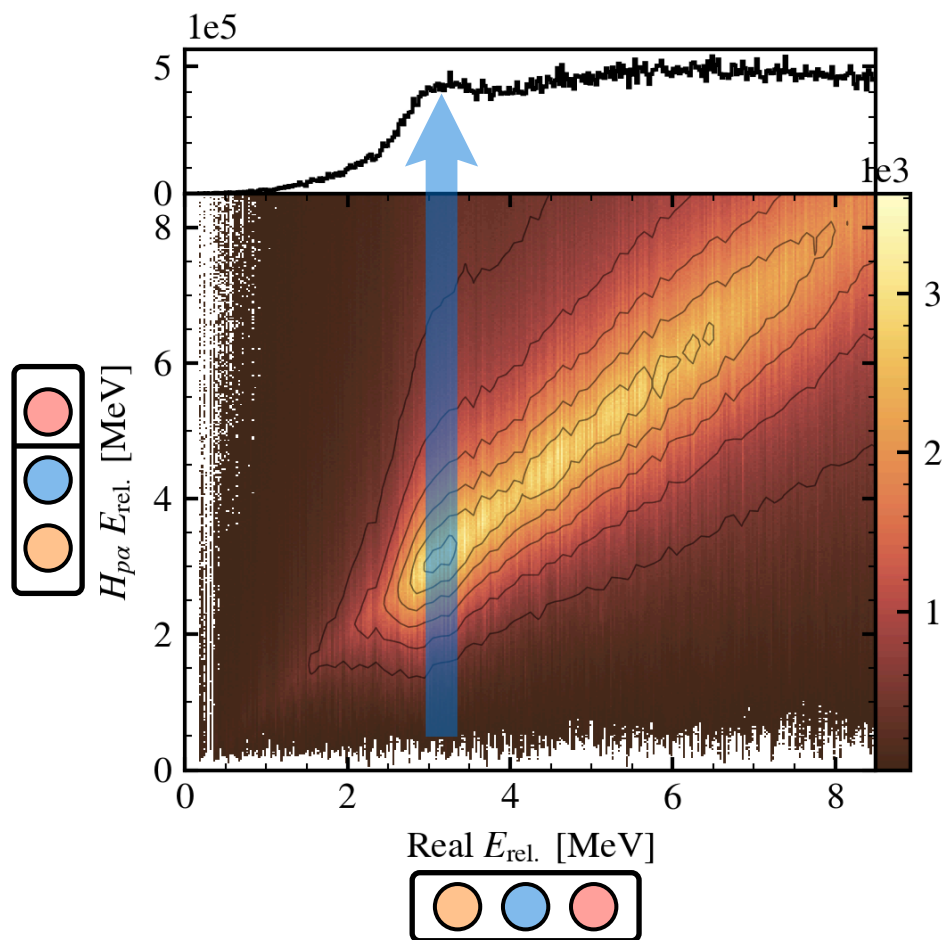
Impact of Real 3-particle Correlations on Mixed Events



Positive Correlation

- Given a particular 3-body E_{rel} :
 - Preserved 2-body correlation is restricted
 - Resulting mixed 3-body E_{rel} is constrained
- Partial Mixing only weakens 3-body correlations

Impact of Real 3-particle Correlations on Mixed Events



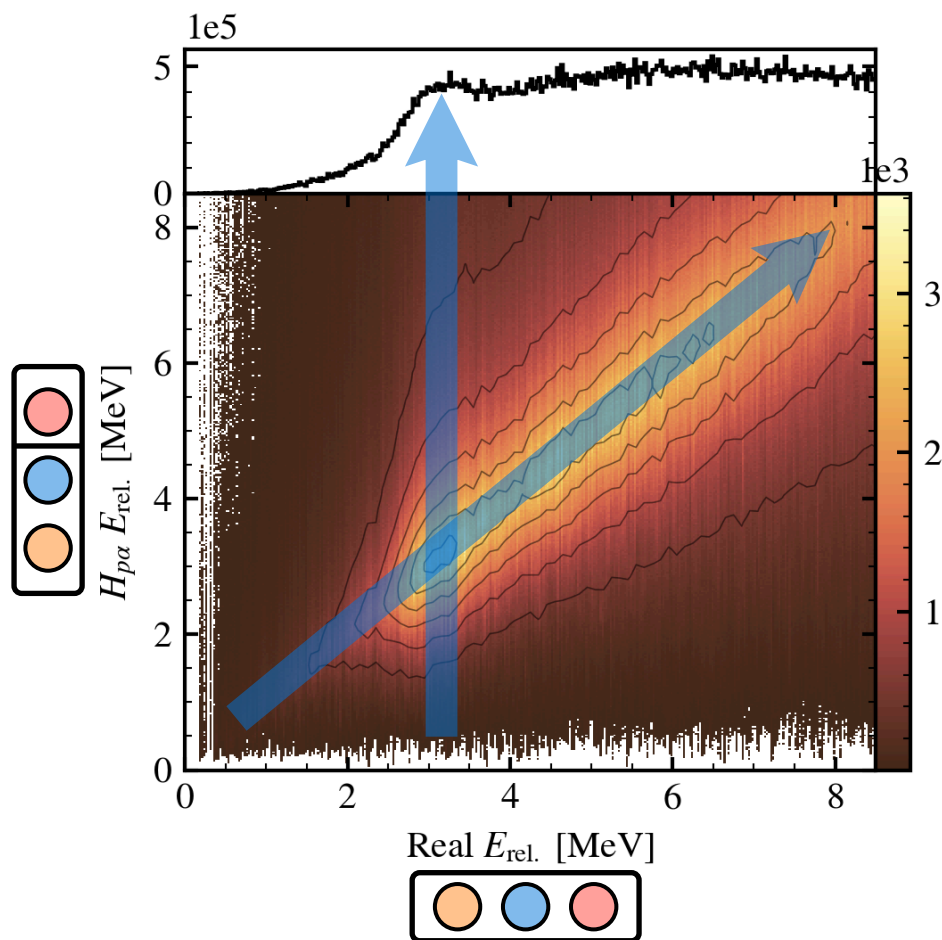
Positive Correlation

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Vertical Correlation

- There is a real 3-body correlation

Impact of Real 3-particle Correlations on Mixed Events



Positive Correlation

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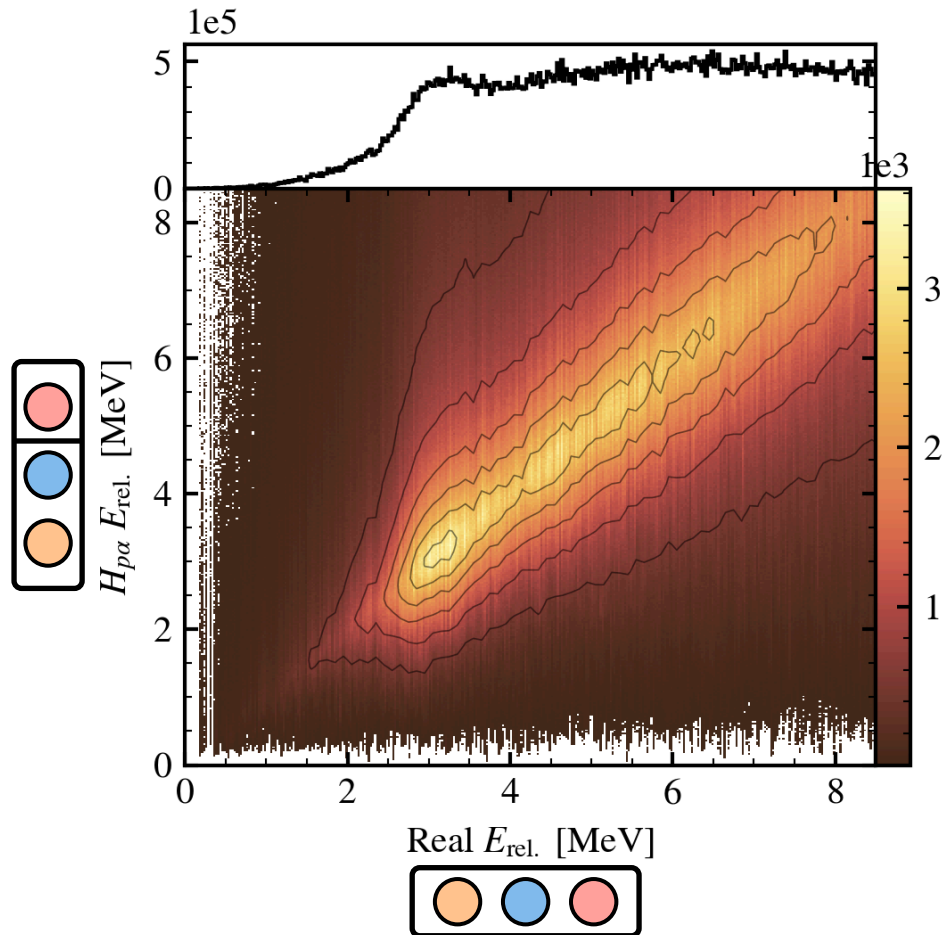
Vertical Correlation

- There is a real 3-body correlation

Combined Effect

- A distortion of the mixed events induced by real correlations

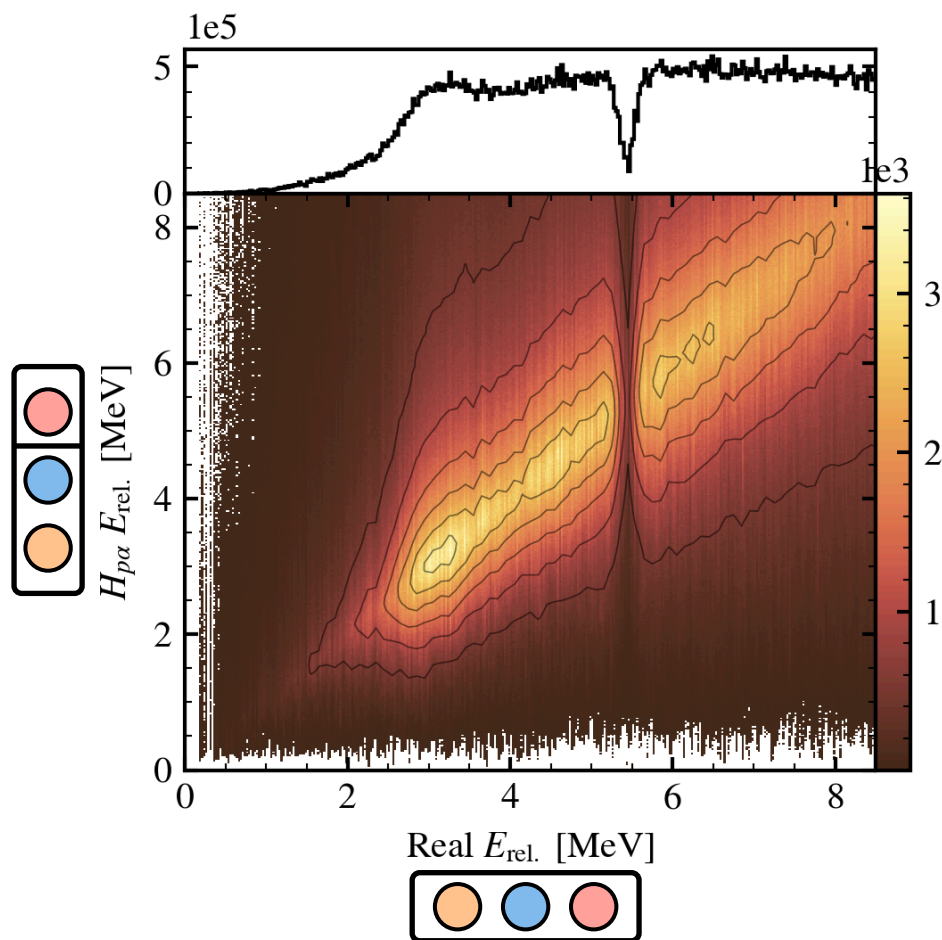
Supression of Signal Overview



Guess the signal

- Subtract it from the real distribution
- Suppress (scale down) the columns in the matrix to match that subtraction

Supression of Signal Overview



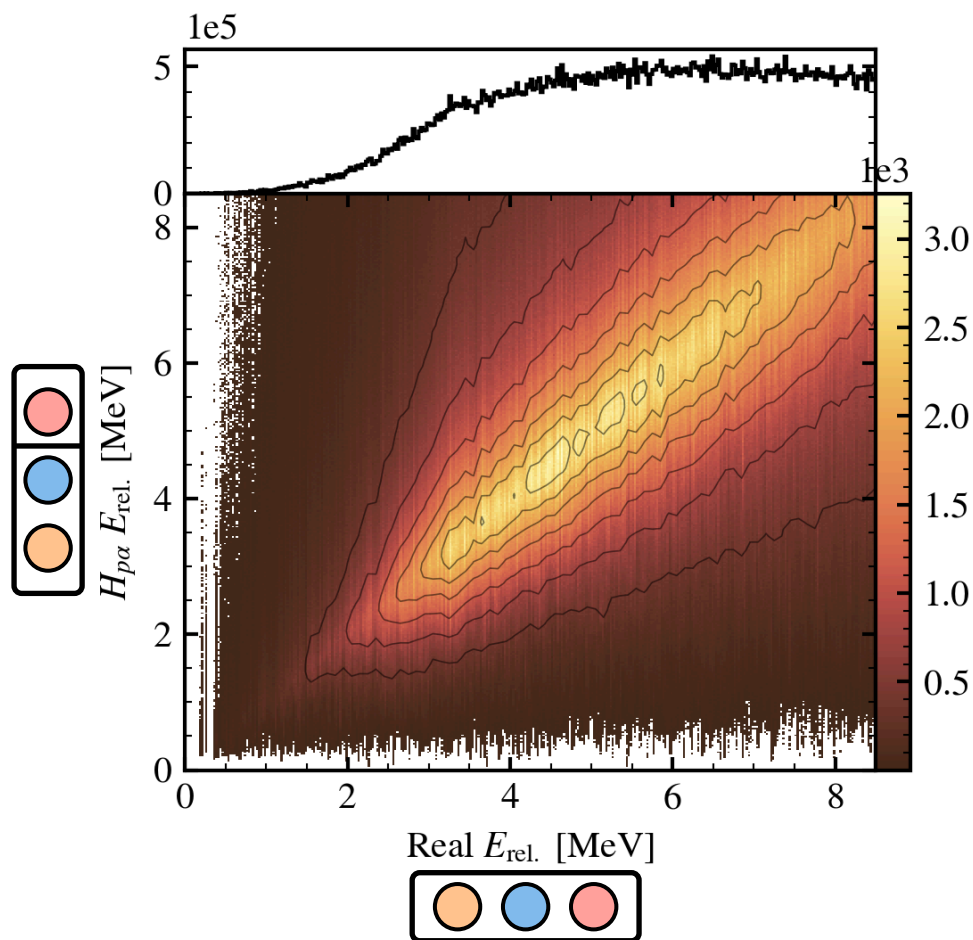
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If Bad Guess

- Effects of the signal on mixed events remain
- Possibly more bias is introduced
- Reconstruction of the background still bad

Supression of Signal Overview



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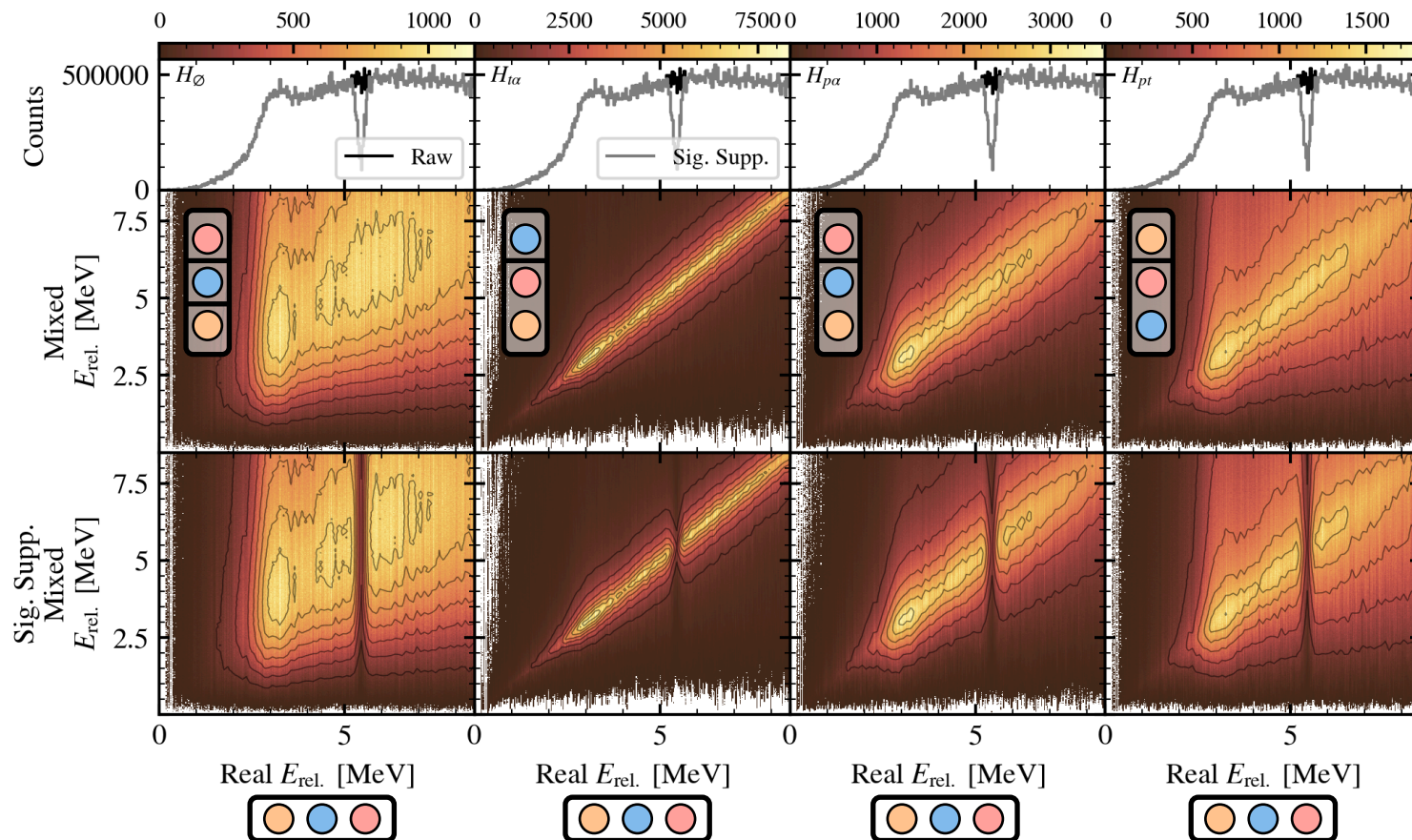
If Good Guess

- Suppress the effects of the signal on mixed events
- Reconstruction of the background is accurate again

Bad Guess Full Example

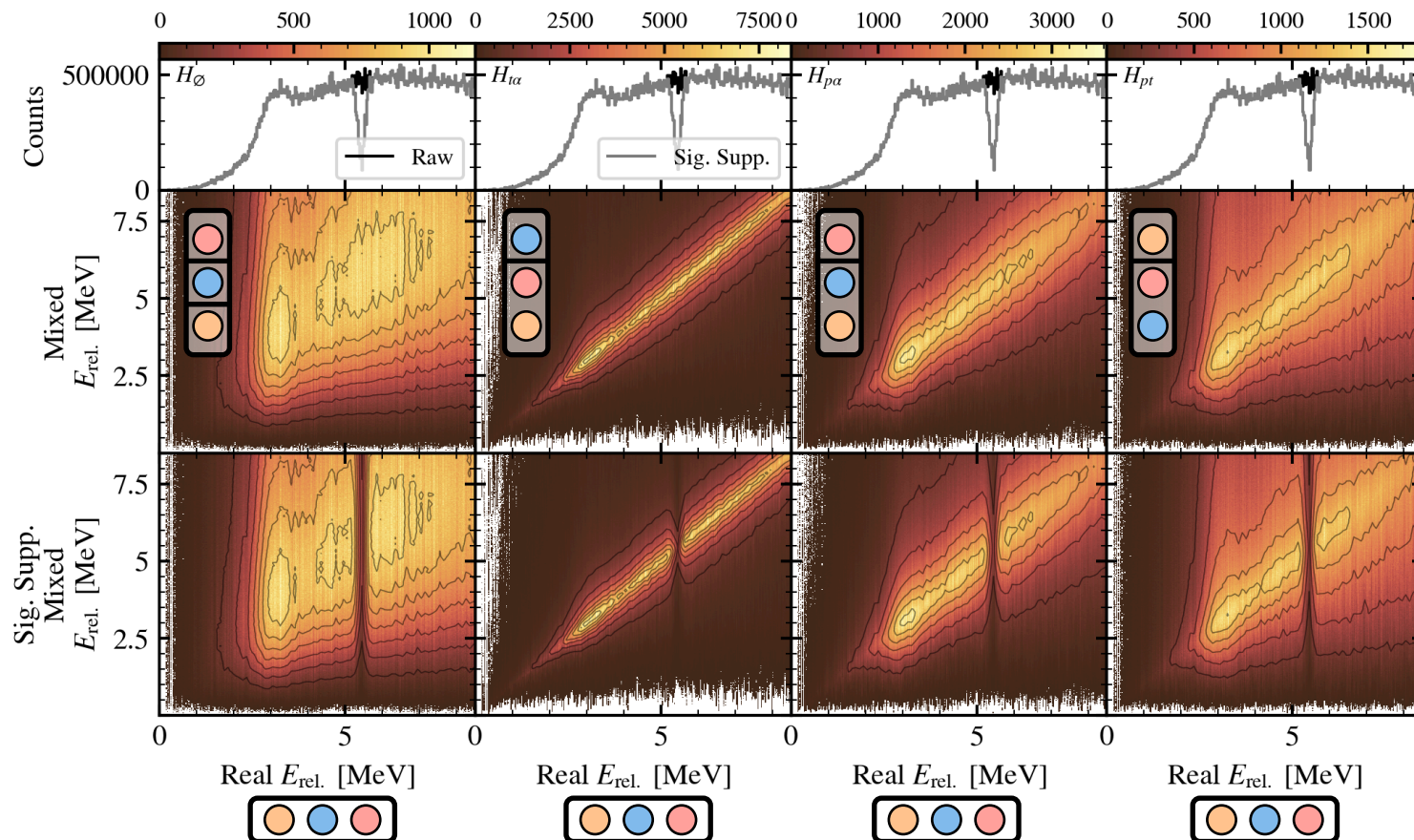
- Assign signal

Bad Guess Full Example

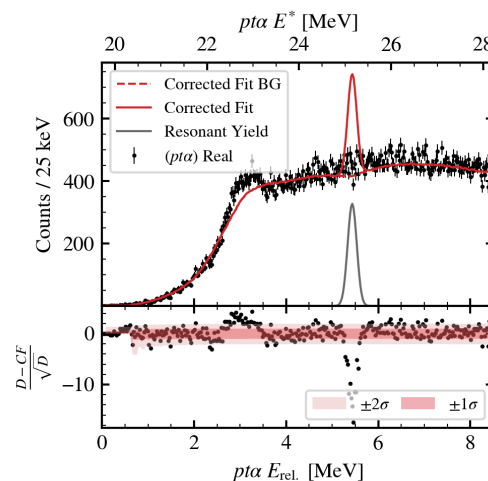


- Assign signal
- Suppress Mixed vs. Real Matrices

Bad Guess Full Example



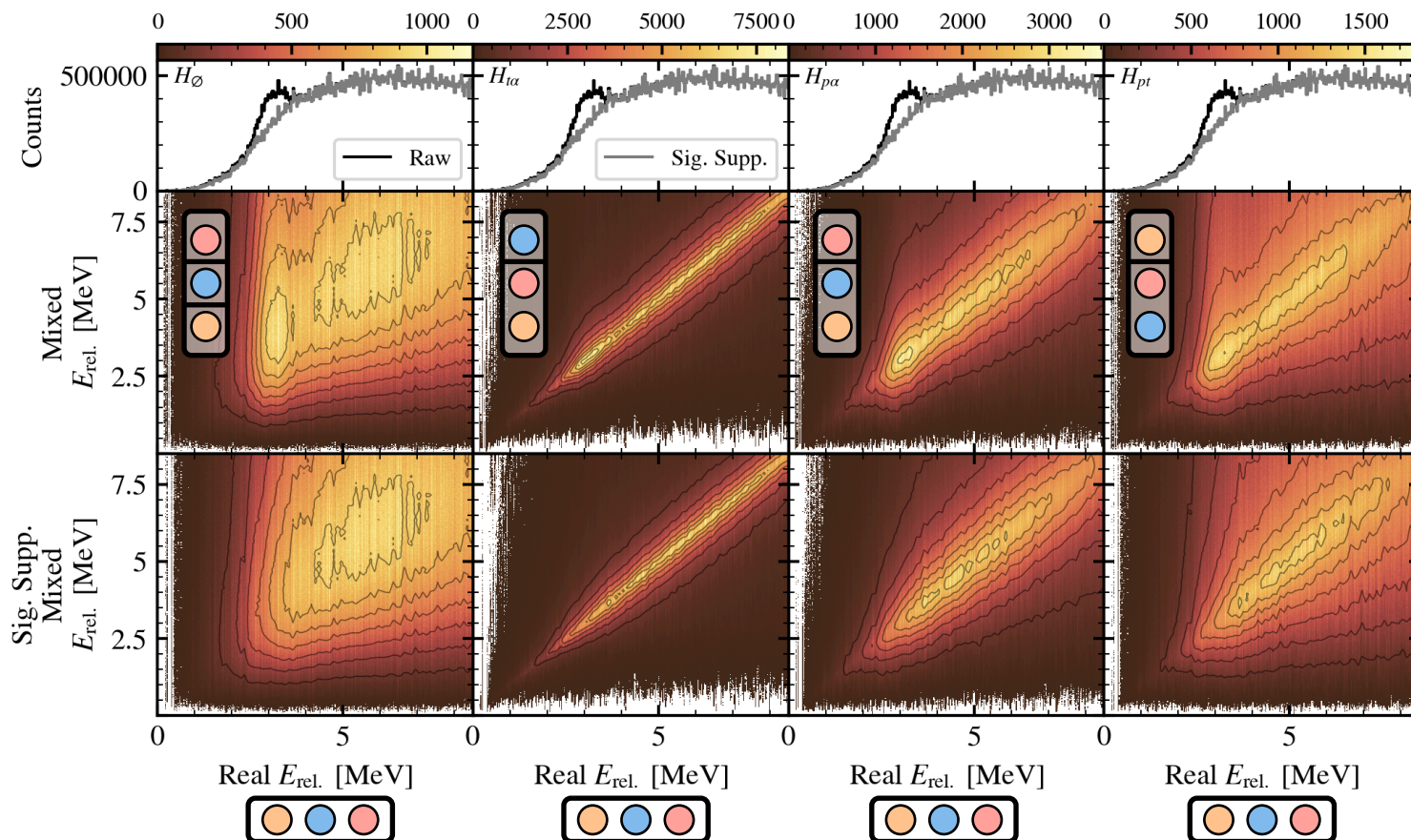
- Assign signal
- Suppress Mixed vs. Real Matrices
- Calculate background



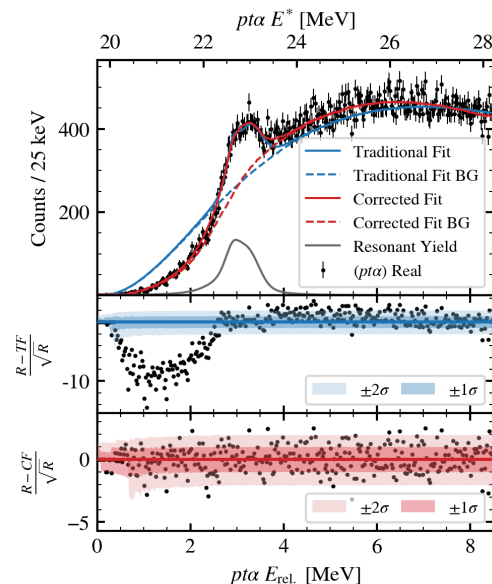
Fit Signal

- **Vary** signal
- Suppress Mixed vs. Real Matrices
- Calculate background
- **Optimize** agreement with measured (max-L)

Fit Signal

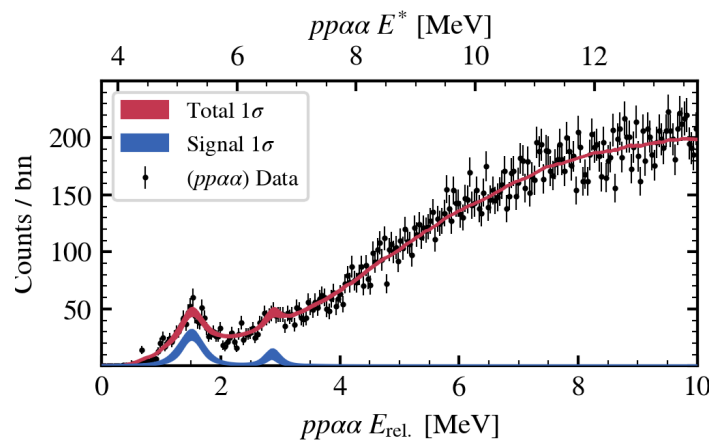
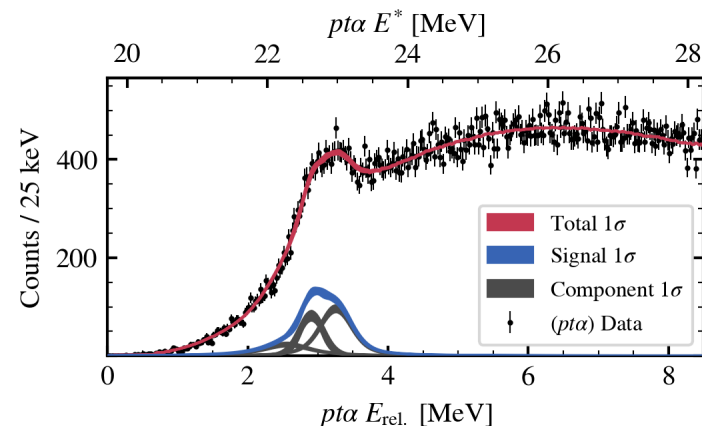
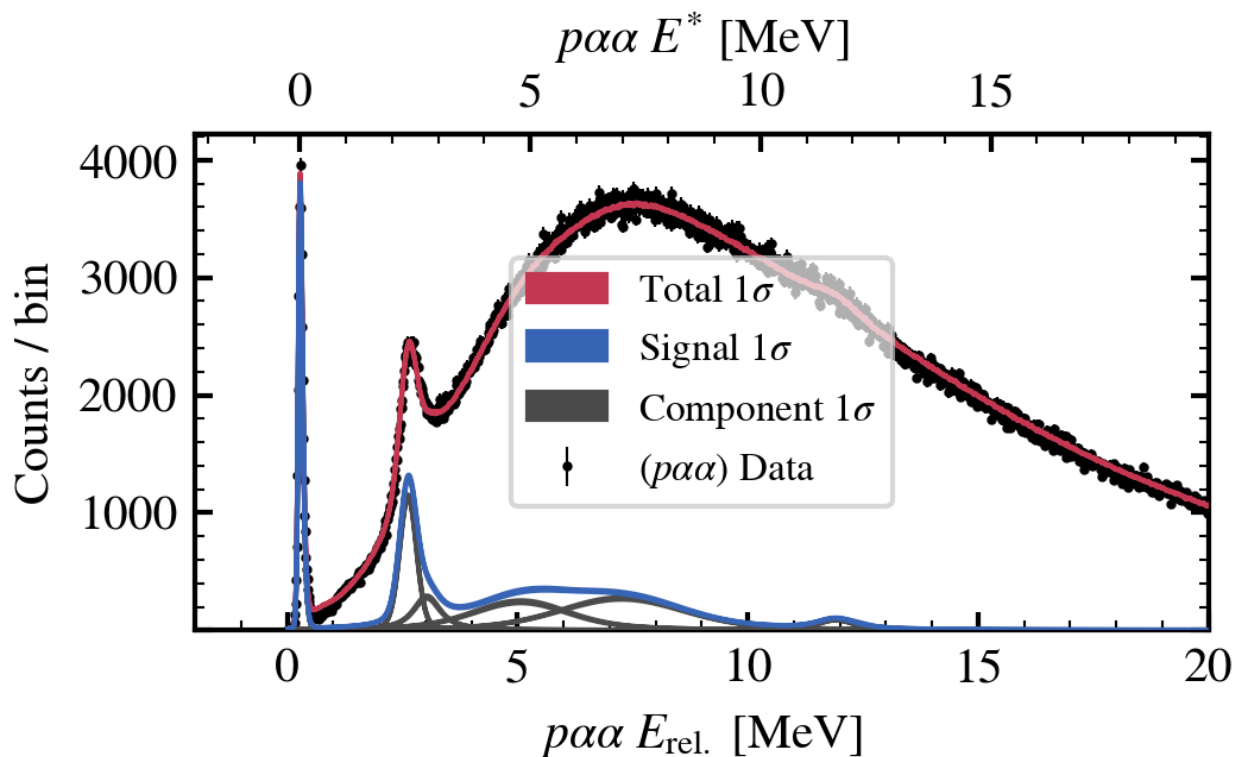


- **Vary** signal
- Suppress Mixed vs. Real Matrices
- Calculate background
- **Optimize** agreement with measured (max-L)



Example Fits

- This technique is successful at extracting subtle resonances from large nonresonant backgrounds
- (all from $^{16}\text{O} + ^{12}\text{C}$ @ 35 MeV/u)



End of Background Method

Application 1: Toroidal Nucleus Search

Semi-Empirical Mass Formula

$$E_B = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(N-Z)^2}{A} \pm \delta(N, Z)$$

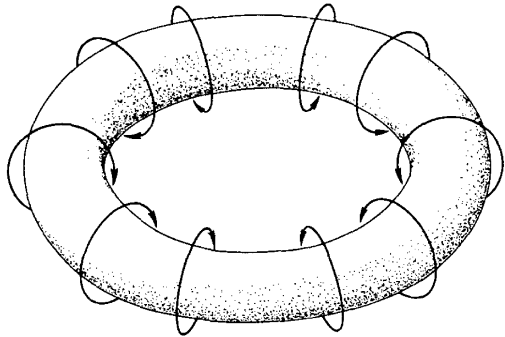
Application 1: Toroidal Nucleus Search

Semi-Empirical Mass Formula

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- E_B depends on **spherical** assumptions.

What happens under deformed toroidal configurations at high angular momentum?



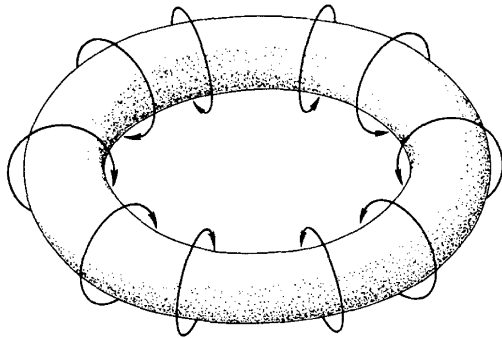
Application 1: Toroidal Nucleus Search

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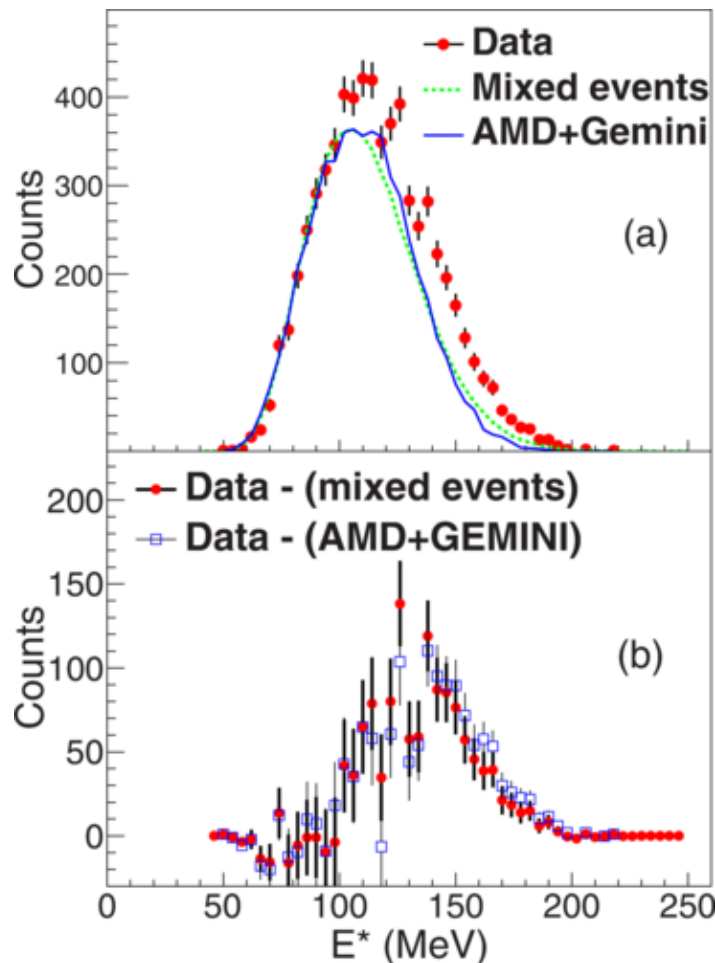
A. Staszczak and C.-Y. Wong predict 18 toroidal isomers

	E^* [MeV]	I [$\hbar$]	d [fm]	$\rho_{\max}/\rho_0$
^{28}Si	143.18	44	1.45	0.74
^{32}S	153.87	48	1.42	0.76
	193.35	66	1.40	0.67
	168.03	56	1.40	0.78
^{36}Ar	198.63	72	1.39	0.71
	238.56	92	1.37	0.64
^{40}Ca	178.36	60	1.40	0.79
	214.23	82	1.39	0.73
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

- Indicators that high energy toroidal states might decay into many α particles

Application 1: Toroidal Nucleus Search

- Reaction
 - $^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV/u
 - Part of 2009 experimental series
- Sample Size
 - 6,467 events w/ 7α
- Resolution for E^* from 7α
 - ~ 9.4 MeV (FWHM) in ROI
 - Position-insensitive detectors

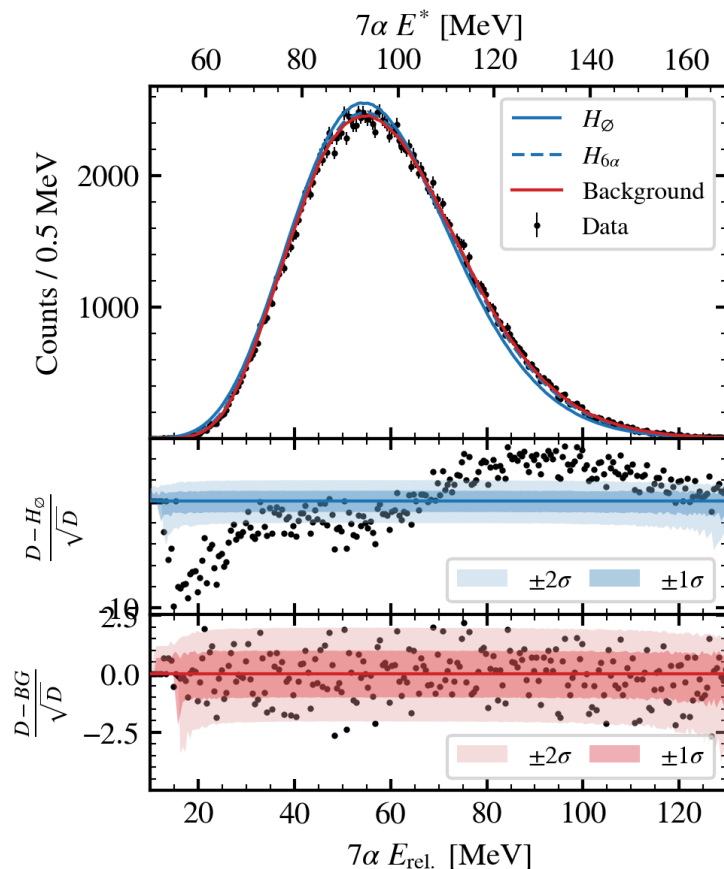


- Nonresonant Background
 - Mixed Events
 - Shifted AMD Distribution
- Extracted Peaks
 - $E^* = \mathbf{114, 126, \& 138}$ MeV
 - FWHM dominated by resolution

“Clearly an experiment with much better angular resolution, allowing better resolution for the excitation energy spectrum, will be very desirable.”

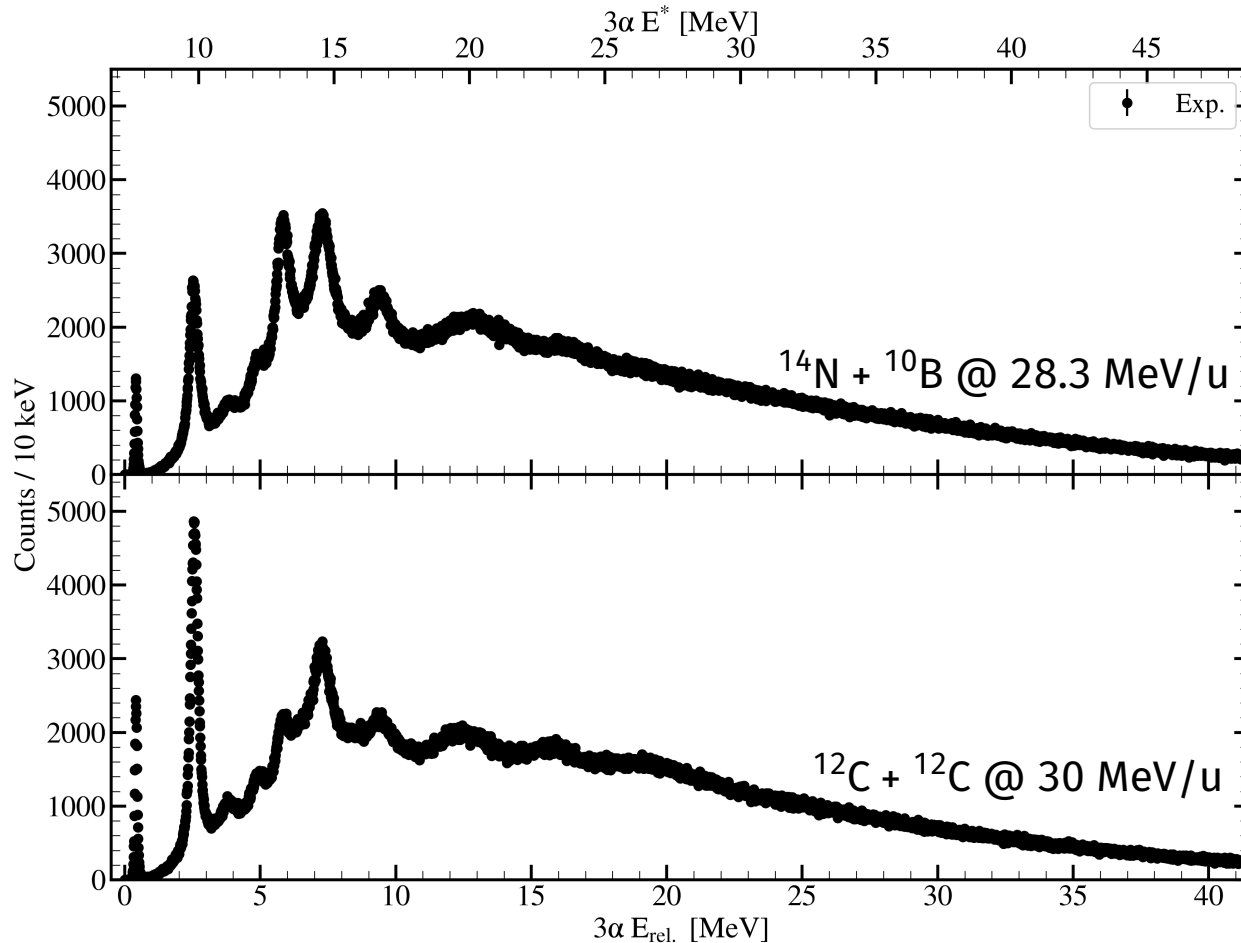
Application 1: Toroidal Nucleus Search

- Reaction
 - $^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV/u
 - Dedicated measurement for toroidal search
- Sample Size
 - 187,067 Events w/ 7α (x29 $\uparrow$)
- Resolution for E^* from 7α
 - ~ 2.5 MeV (FWHM) in ROI (x3.5 $\downarrow$)



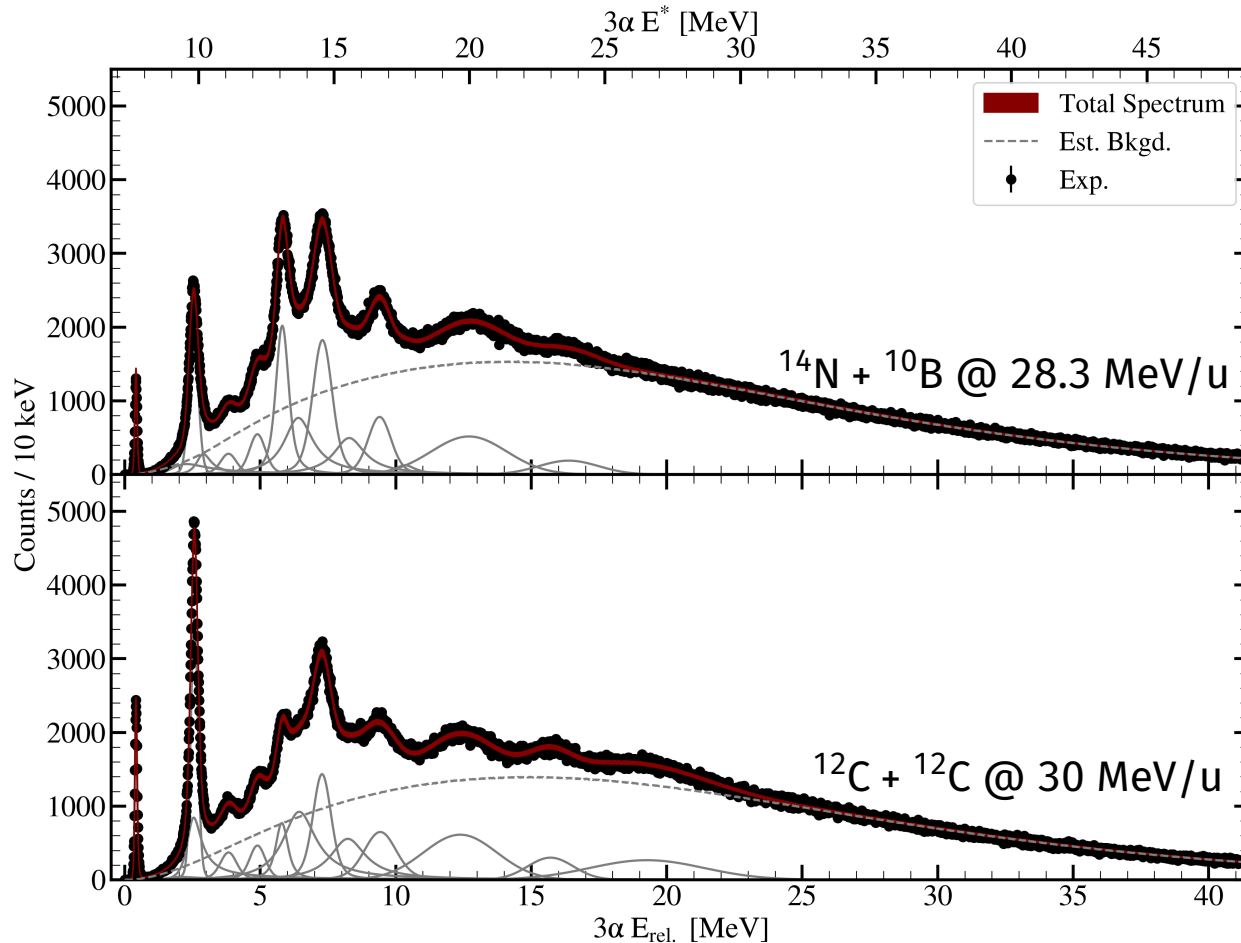
- Far more sensitive measurement
- Spectrum is statistically consistent with a no-signal model
- Active work is
 - Setting upper limits on production
 - Extending the search to other systems

Application 2: Reaction-Dependent State Population (T. Hankins)



- Strong alpha clustering can influence the dynamics of nuclear reactions.
- Seek to understand through relative population rates between two reactions
 - (e.g., $^{12}\text{C}^* \rightarrow 3\alpha$)

Application 2: Reaction-Dependent State Population (T. Hankins)



- Fits using proper lineshape descriptions yield very good total spectrum characterizations.
 - Contributions at $E_{\text{rel}} < 10$ MeV are real states known to decay $^{12}\text{C}^* \rightarrow 3\alpha$

Acknowledgments



- **SJY Group**

- Travis Hankins
- Andy Hannaman (former)
- Alan McIntosh
- Kris Hagel
- Sherry Yennello
- *et al.*



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- Department of Energy: DE-FG02-93ER40773

Questions and Comments Welcome

